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Ward et al.

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(54) **SYSTEM AND METHOD FOR ASSEMBLING AND/OR OPERATING CONTROL SWITCH IN RELATION TO WASTE DISPOSER AND IN RELATION TO ALTERNATIVE ELECTRIC POWER SOURCES**

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B02C 18/00 (2006.01)
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CPC ***E03C 1/2665*** (2013.01); ***B02C 18/0092*** (2013.01); ***H01R 13/639*** (2013.01); ***H02K 5/225*** (2013.01); *B02C 23/36* (2013.01)

(58) **Field of Classification Search**
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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,244,402	A	6/1941	Powers
2,477,686	A	8/1949	Coss

(Continued)

FOREIGN PATENT DOCUMENTS

CN	204753748	11/2015
CN	205617508	10/2016

(Continued)

OTHER PUBLICATIONS

International Search report and Written Opinion for International Application No. PCT/US2023/029927 from WIPO dated Nov. 30, 2023 (15 pages).

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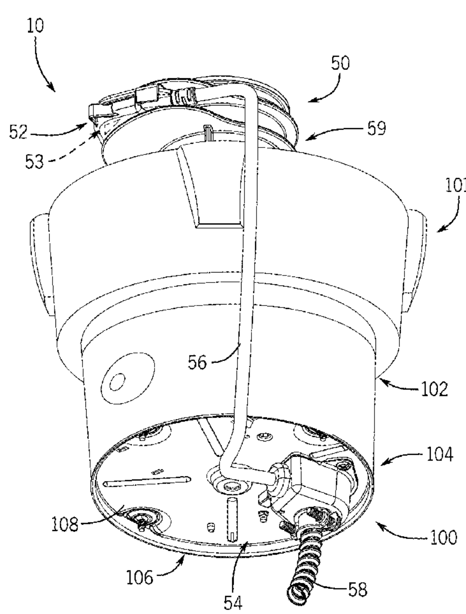
(57) **ABSTRACT**

Waste disposer systems and cover switch mechanisms for implementation in such systems, and related methods, are disclosed. In an example embodiment, a cover switch mechanism includes a cord, a body configured to be coupled to a top portion of a disposer, and connecting components. The connecting components include a terminal cover configured to be coupled to a bottom portion of the disposer, a switch interface connector, and a connection component. The cord extends between the body and cover, where an end of the cord extending into the cover includes first and second leads. The first lead is coupled to the interface connector, and the cover is configured to receive a second end of a power link. The interface connector is configured to be coupled to a third lead of the link, and the connection component is configured to couple the second lead with a fourth lead of the link.

18 Claims, 17 Drawing Sheets



(51) **Int. Cl.**

B02C 23/36	(2006.01)
H01R 13/639	(2006.01)
H02K 5/22	(2006.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,579,400	A	12/1951	Schindler	
2,642,232	A *	6/1953	Drew	E03C 1/2665 241/46.016
2,657,397	A *	11/1953	Drew	E03C 1/2665 241/46.016
2,850,244	A *	9/1958	James	E03C 1/2665 241/32.5
2,860,834	A	11/1958	Hammes	
2,961,172	A	11/1960	Wieczorek et al.	
2,979,274	A *	4/1961	Tull	E03C 1/2665 285/31
3,425,637	A	2/1969	Enright et al.	
3,464,638	A	9/1969	Enright et al.	
3,504,863	A	4/1970	Burkland et al.	
6,082,643	A	7/2000	Kovacs	
7,503,514	B2	3/2009	Berger et al.	
7,757,981	B2	7/2010	Anderson et al.	
9,145,666	B2	9/2015	Hammer	
9,968,940	B2	5/2018	Hartmann et al.	
10,981,178	B2	4/2021	Britto	
11,441,302	B2	9/2022	Reidel et al.	
11,532,909	B2 *	12/2022	Weaver	H01R 4/48
2004/0178289	A1	9/2004	Jara-Almonte et al.	
2006/0038047	A1	2/2006	Anderson et al.	
2021/0087801	A1 *	3/2021	Ward	H01R 13/73
2021/0238834	A1 *	8/2021	Ward	H01R 25/006
2022/0120069	A1 *	4/2022	Reidel	B02C 18/0092

FOREIGN PATENT DOCUMENTS

GB	2392636	B	3/2004
JP	4968711	B2	7/2012
WO	2007056631	A2	5/2007

* cited by examiner

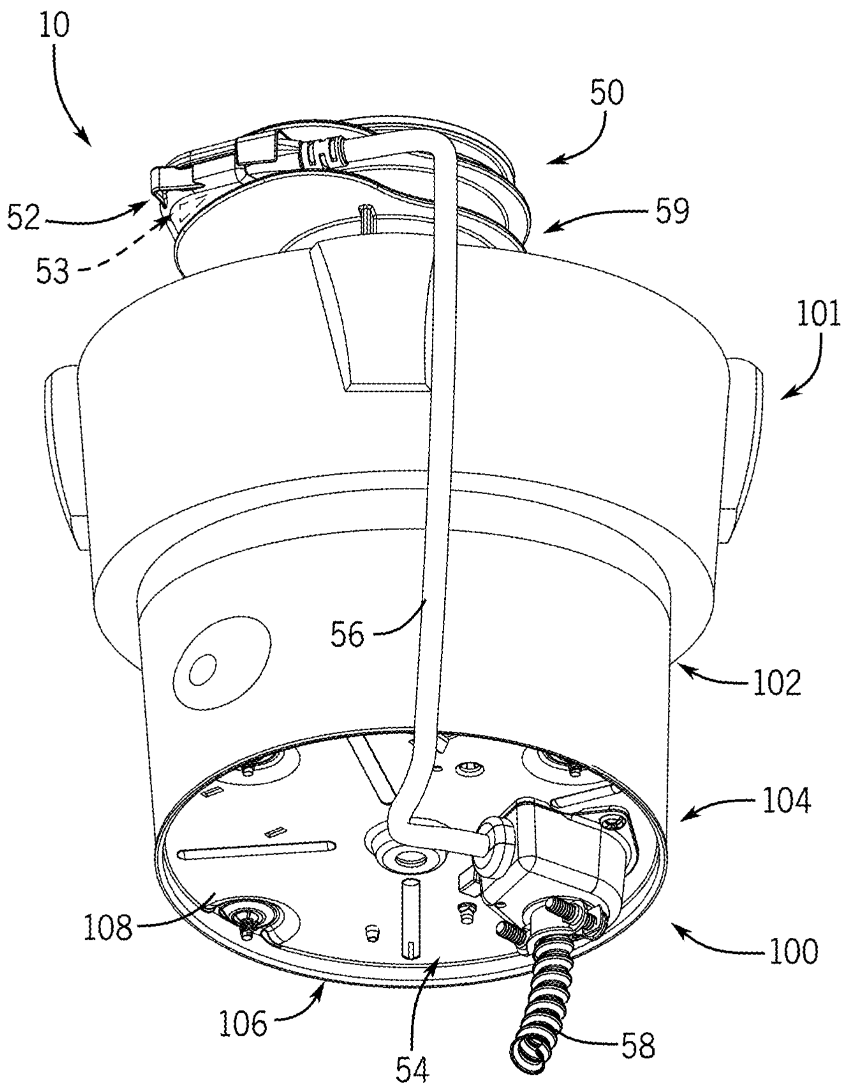


FIG. 1

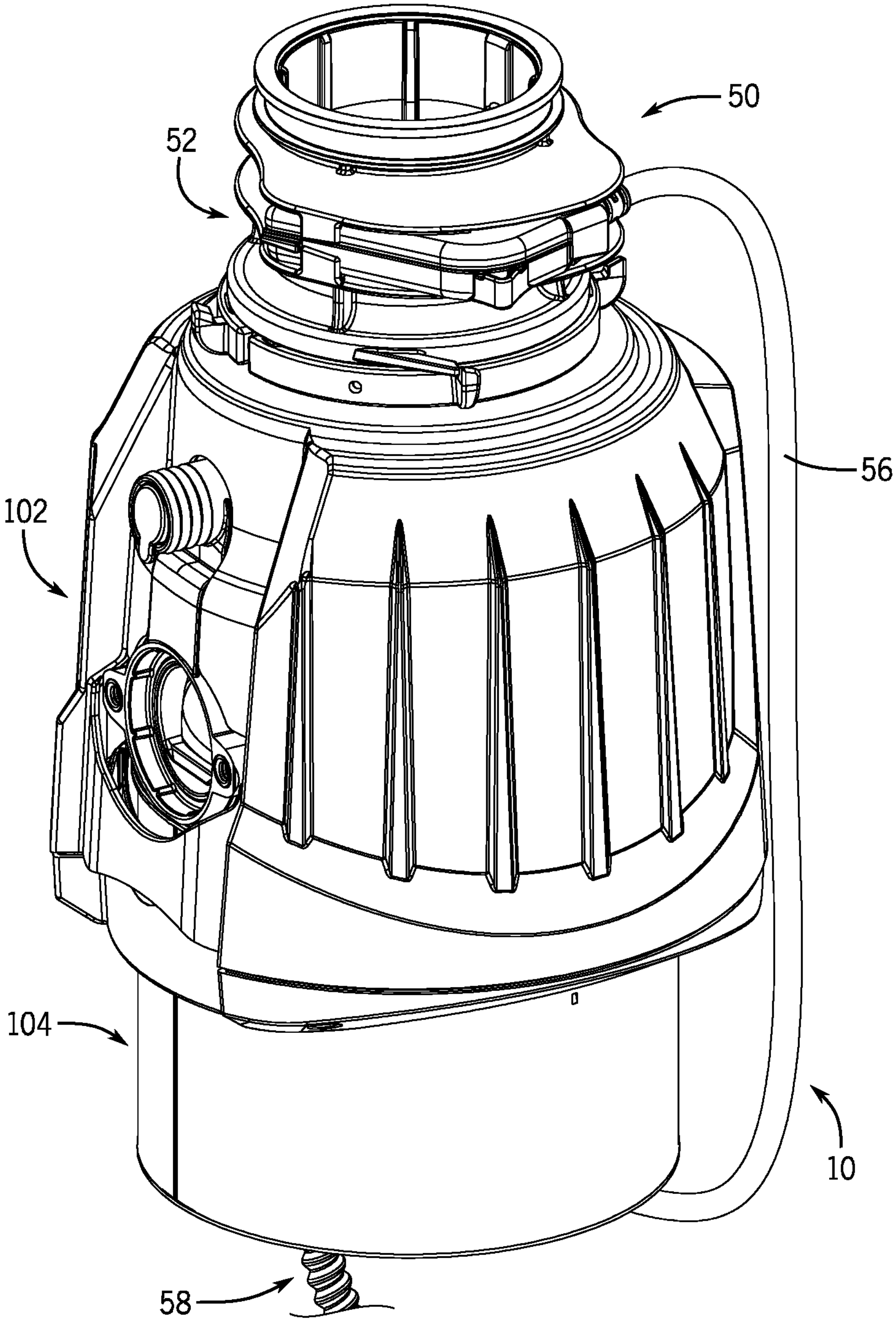


FIG. 2

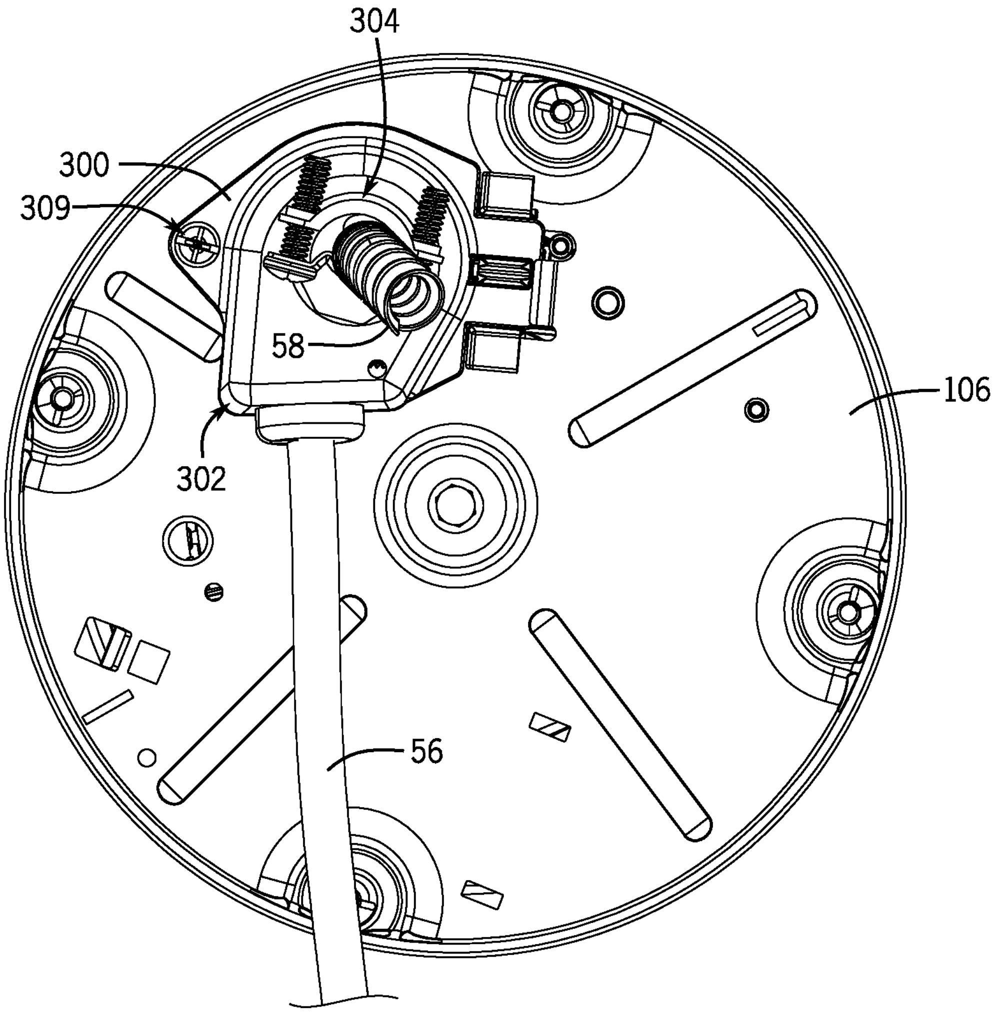


FIG. 3

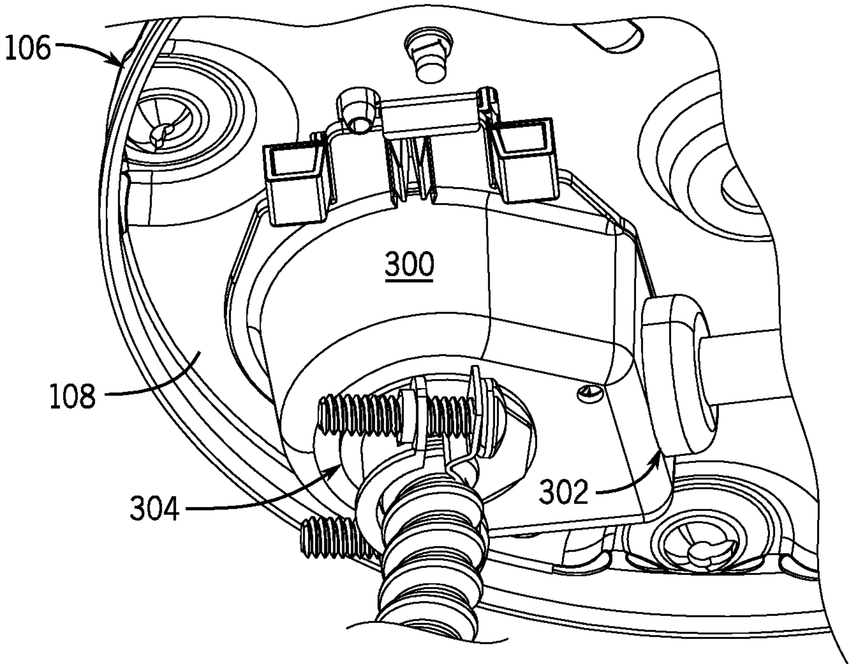


FIG. 4

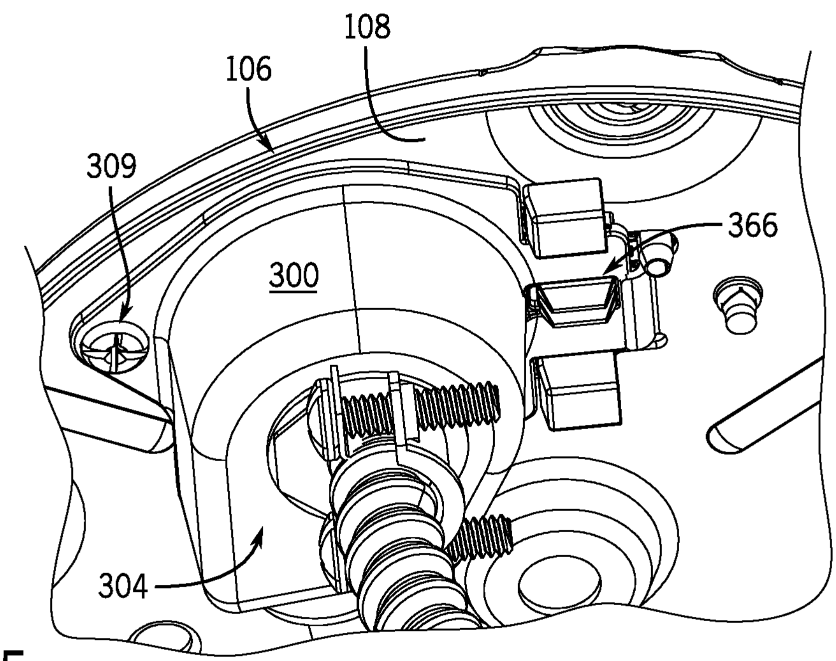


FIG. 5

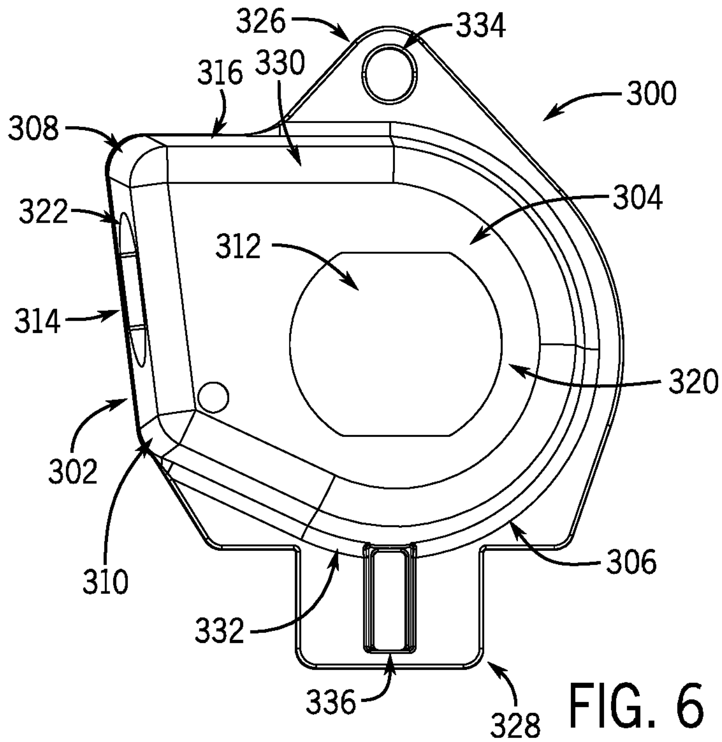

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FIG. 6

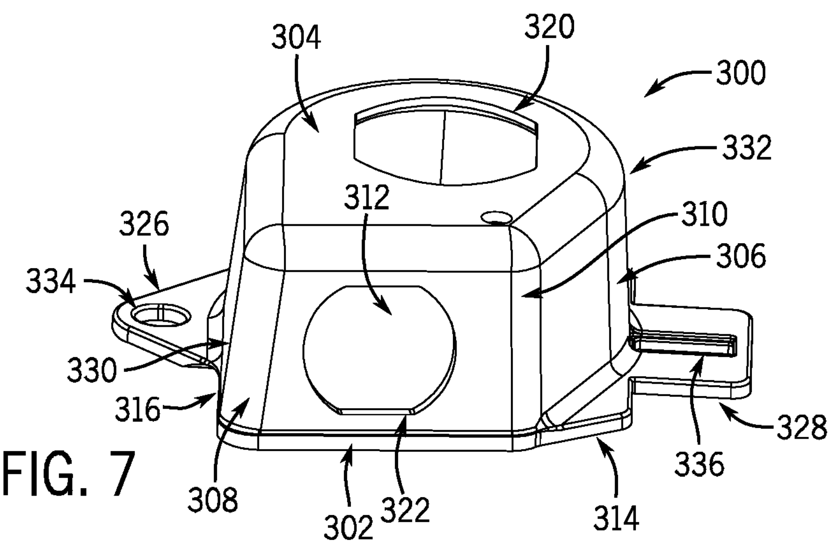
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FIG. 7

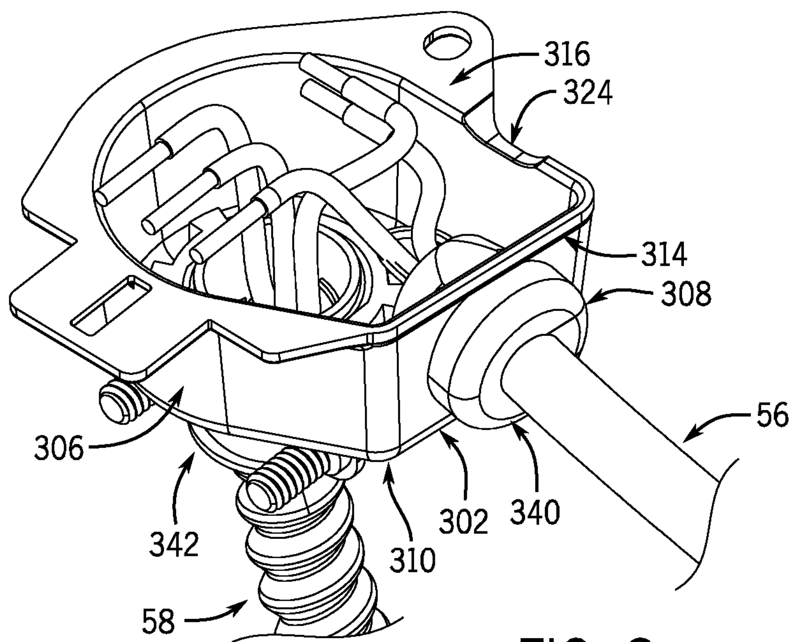

FIG. 8

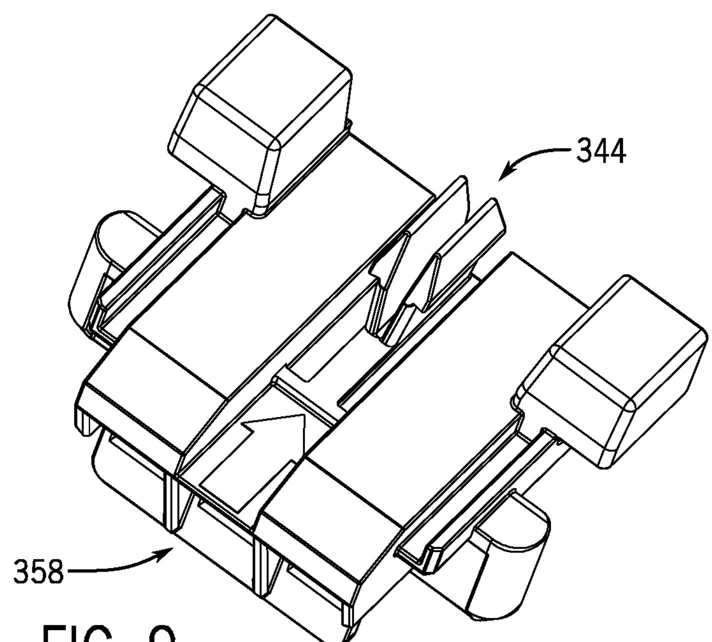

FIG. 9

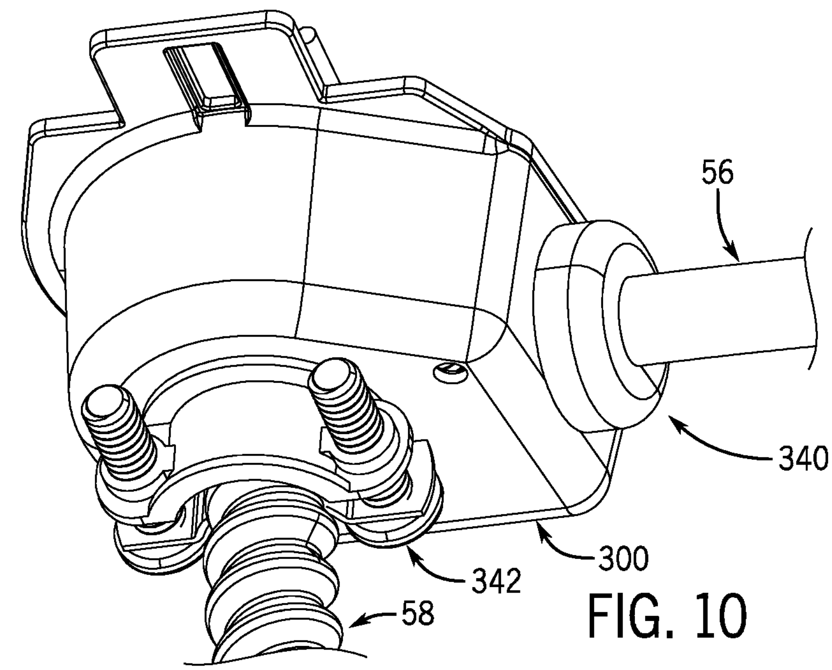

FIG. 10

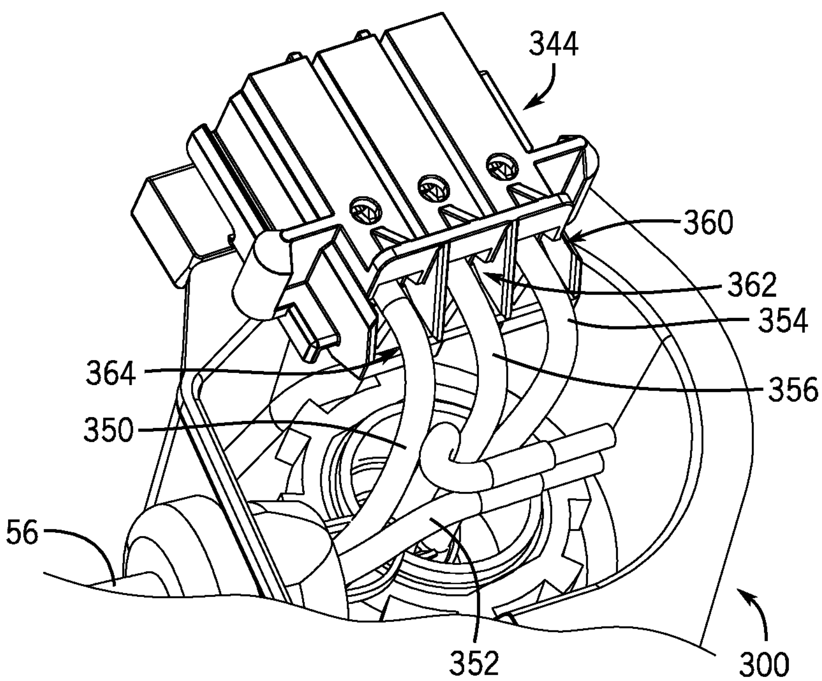

FIG. 11

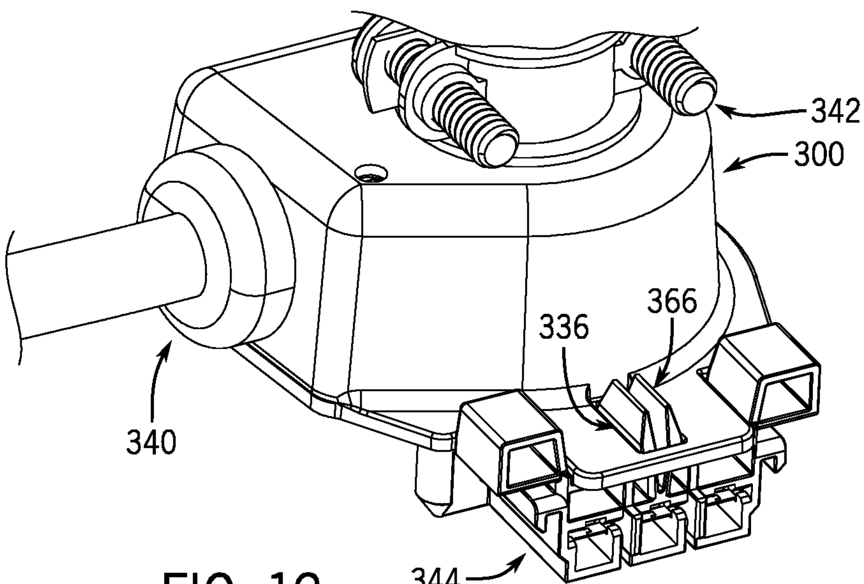

FIG. 12

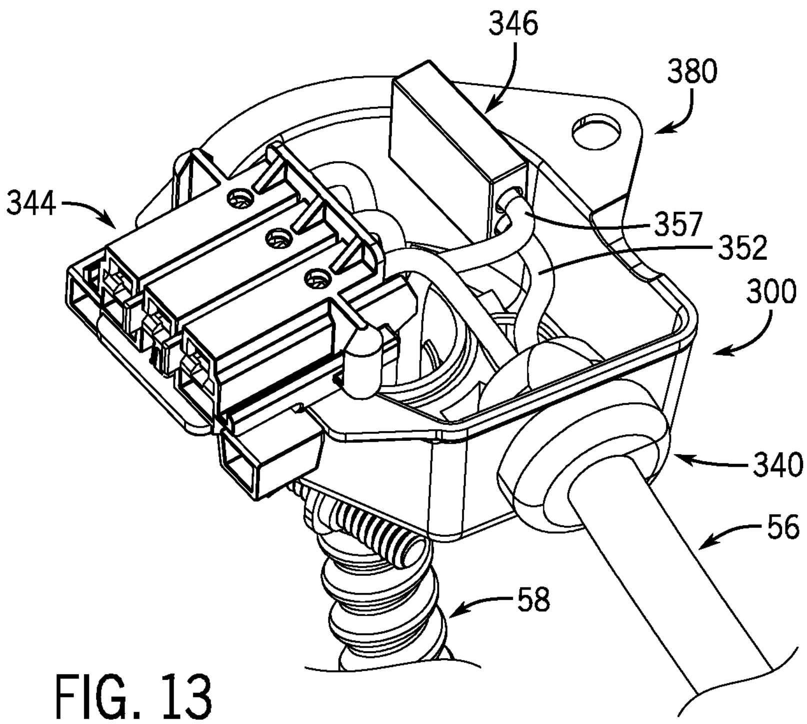

FIG. 13

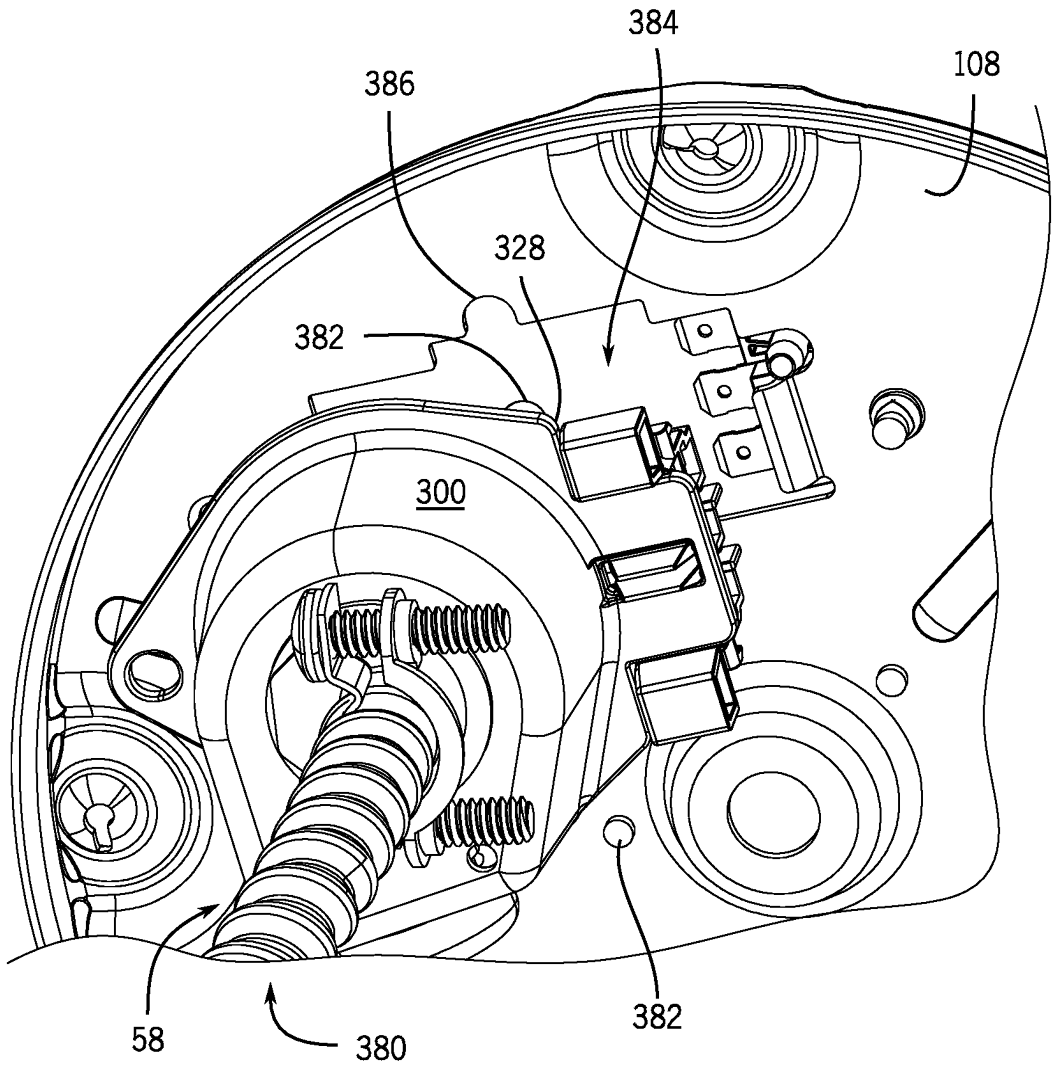

FIG. 14

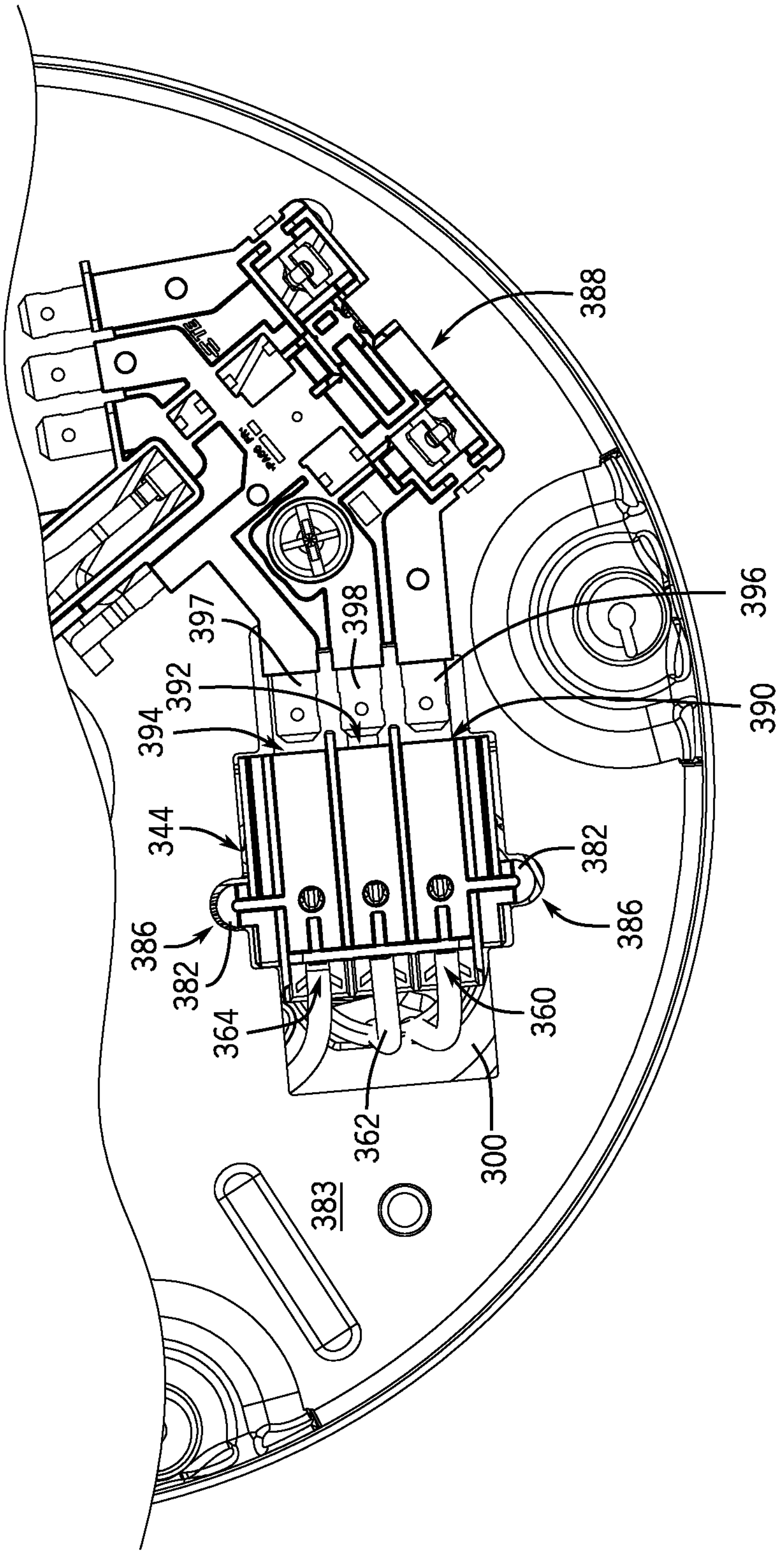
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FIG. 15

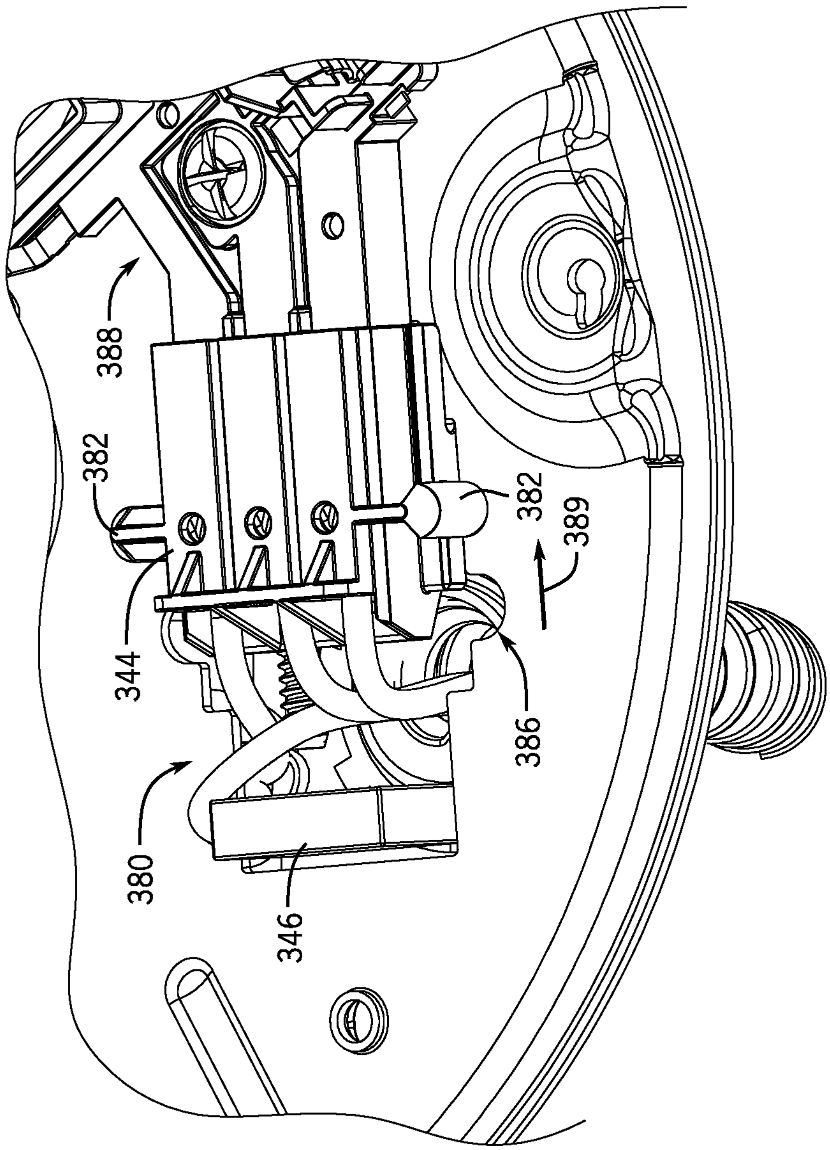

FIG. 16

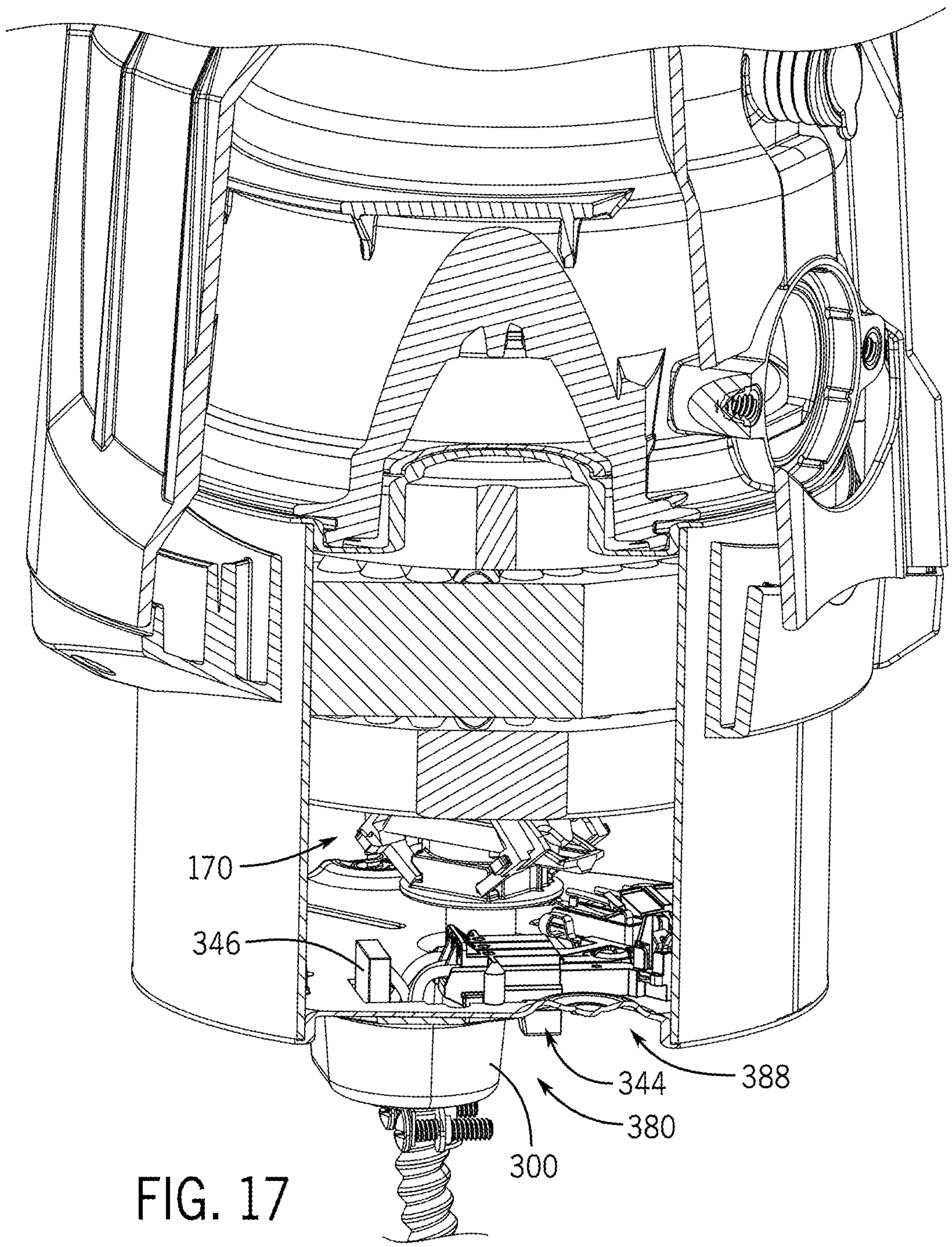

FIG. 17

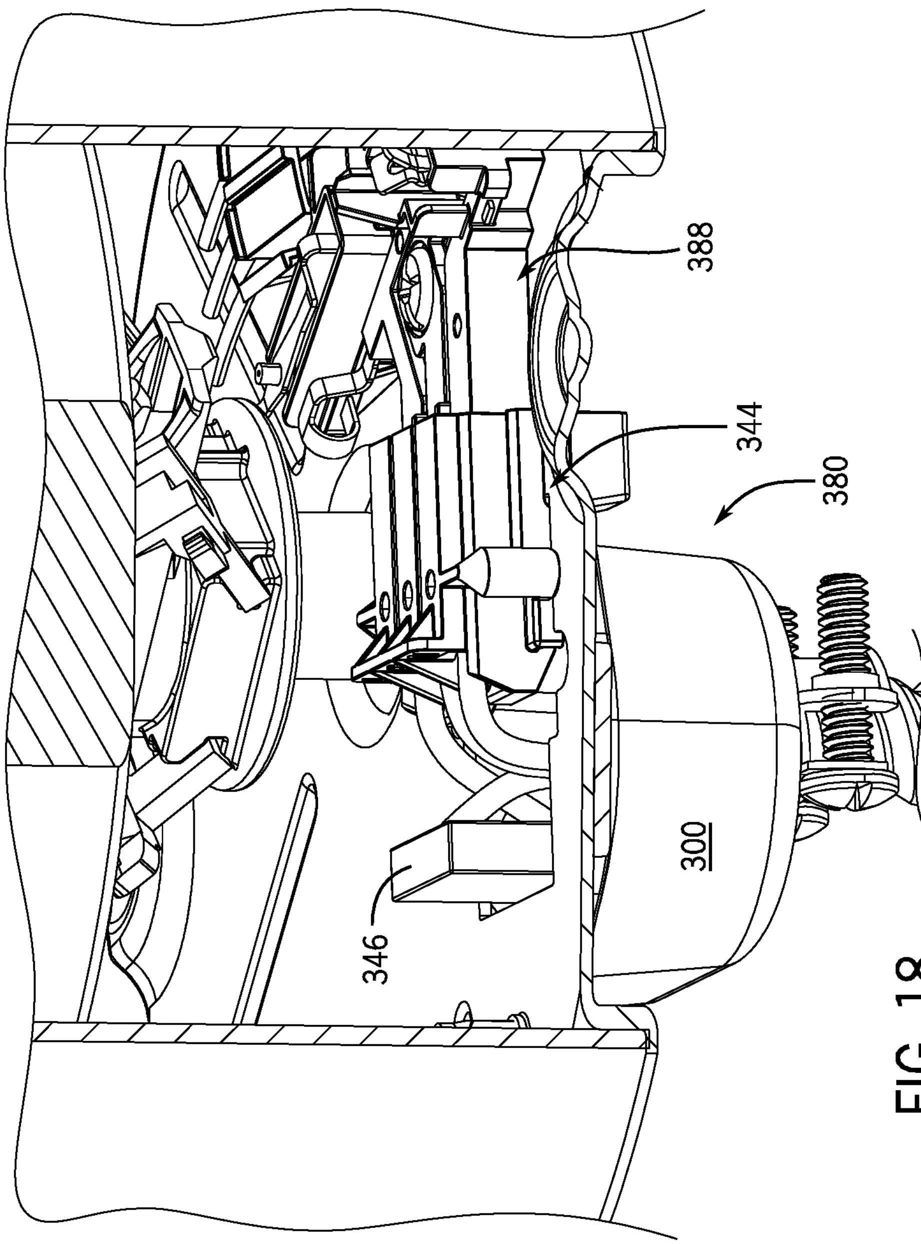

FIG. 18

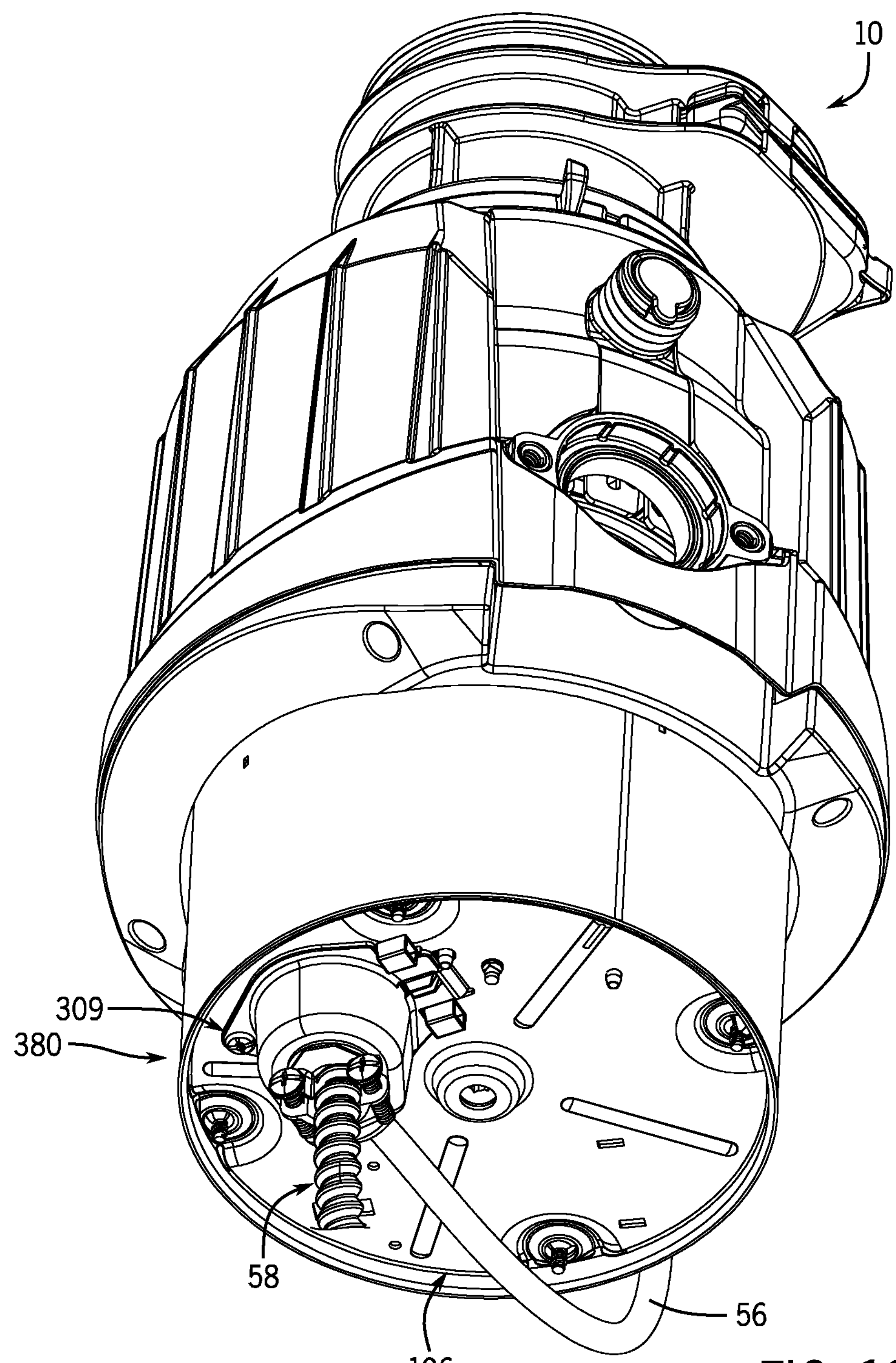

FIG. 19

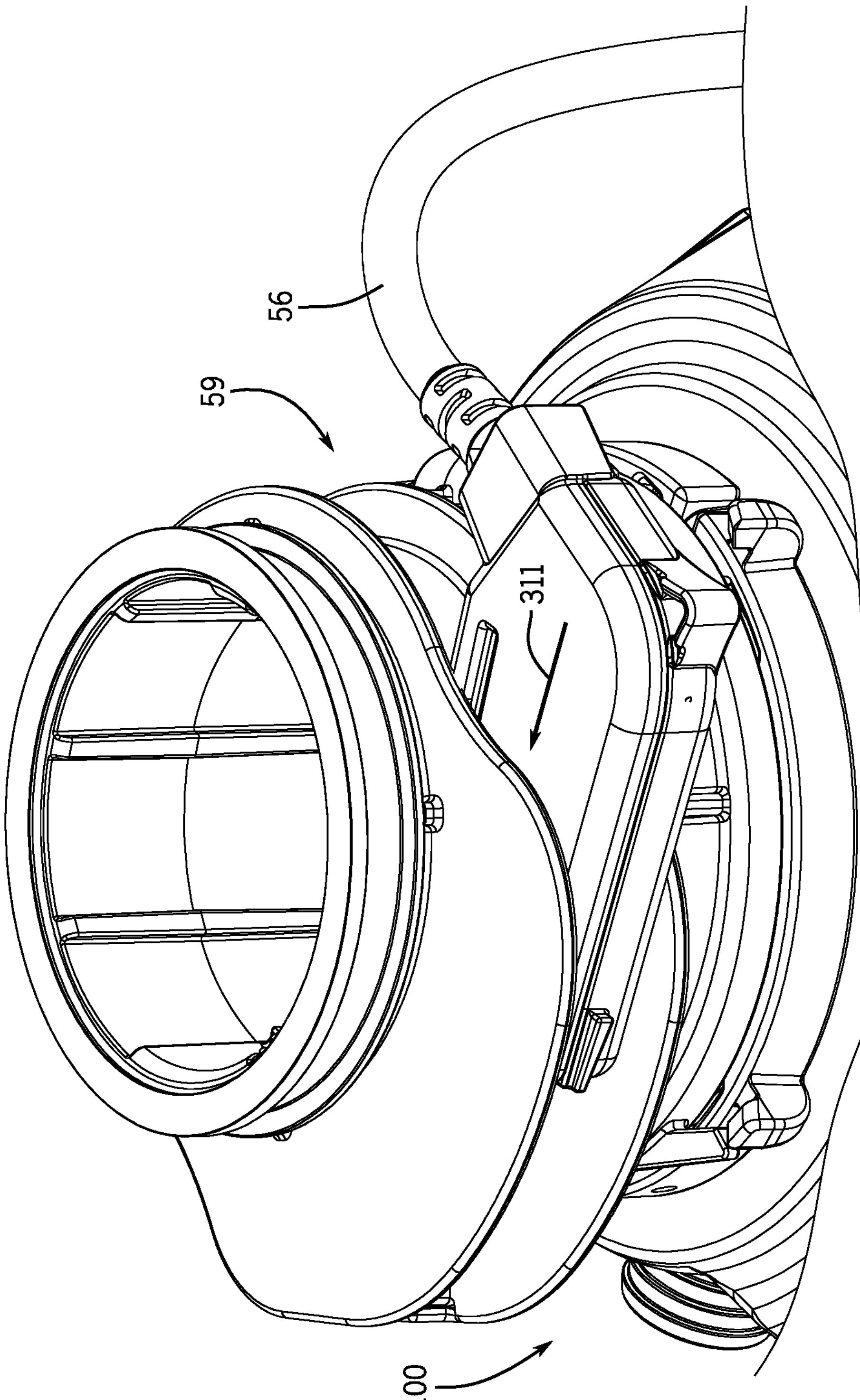

FIG. 20

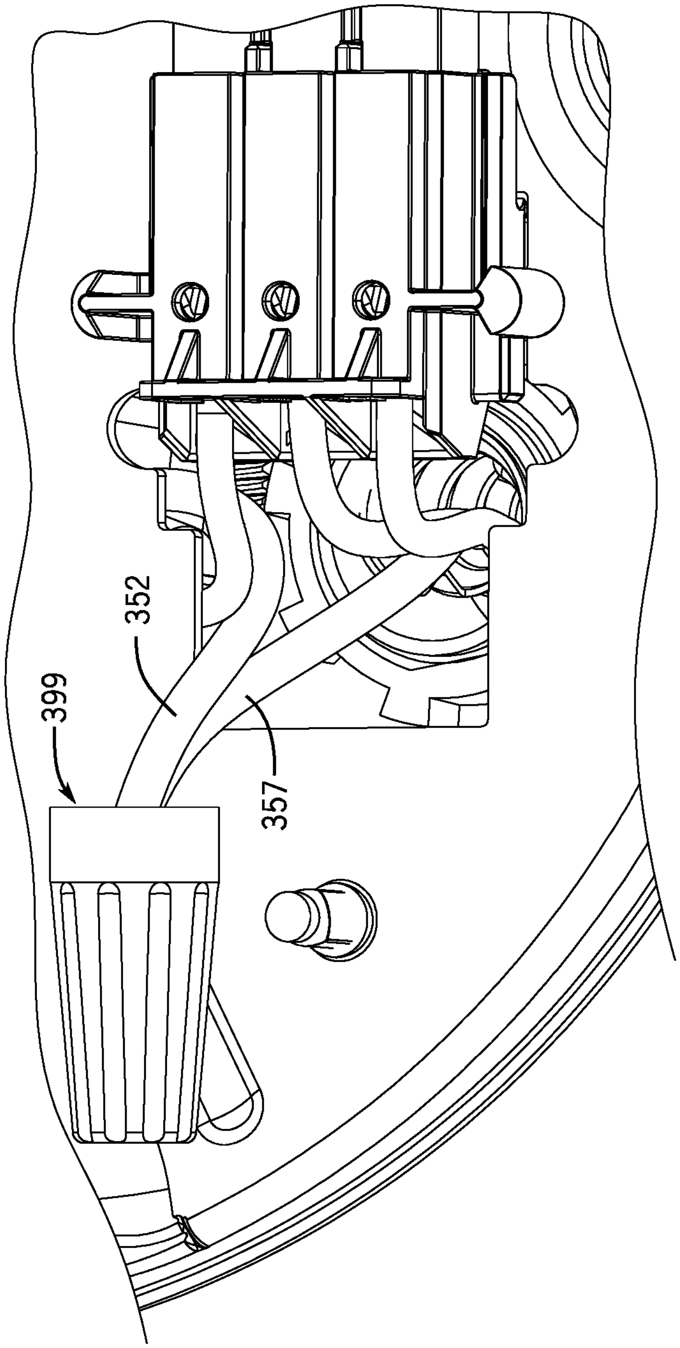

FIG. 21

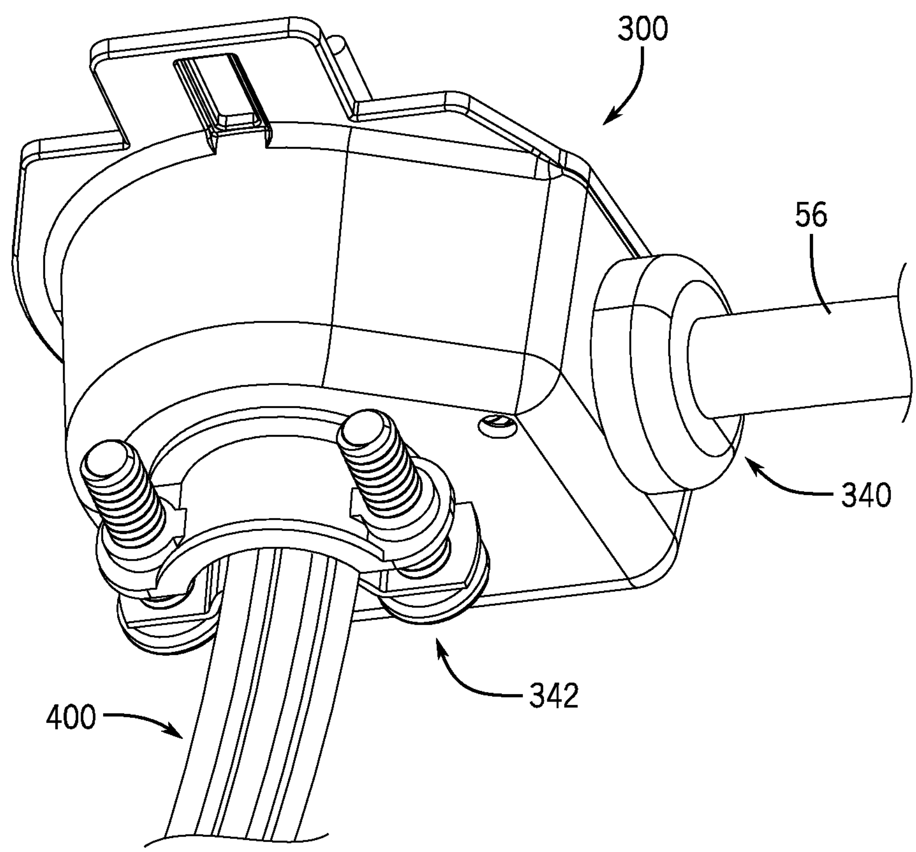

FIG. 22

SYSTEM AND METHOD FOR ASSEMBLING AND/OR OPERATING CONTROL SWITCH IN RELATION TO WASTE DISPOSER AND IN RELATION TO ALTERNATIVE ELECTRIC POWER SOURCES

FIELD

The present disclosure relates to waste disposers such as food waste disposers and, more particularly, to control systems for use in or in conjunction with such waste disposers, as well as to waste disposers comprising such control systems, and to methods of assembling and/or operating control systems in relation to waste disposers.

BACKGROUND

Food waste disposers are used to comminute food scraps into particles small enough to pass through household drain plumbing. Some food waste disposers have operator-actuatable switching mechanisms in which the operator-actuatable switch is positioned at or near the top of the food waste disposer. Because the operator-actuatable switch is positioned at or near the top of the food waste disposer, a wire cable typically is provided that extends from the operator-actuatable switch to (or substantially to) the bottom of the food waste disposer, at which is located a motor (and possibly related control circuitry such as a start switch).

Although helpful for facilitating an operator's accessing of the switching mechanism, such conventional arrangements of operator-actuatable switching mechanisms are limiting in several respects. In particular, such conventional arrangements are typically highly specialized in that the arrangements are respectively configured to be implemented only in relation to respective types of food waste disposers, such that any given arrangement only can be implemented in regard to a particular food waste disposer model or type.

Further, such conventional arrangements of operator-actuatable switching mechanisms are typically configured for implementation in regard to only a single type of home installation circumstance. It will be appreciated that most newer homes have a standard electrical power outlet (e.g., a wall outlet) near the disposer, to which the disposer can be coupled so as to receive power, but nevertheless most older homes (pre-1970's) have a Romex/BX cable extending from a wall of the home, which can be coupled and terminated directly to the disposer in a hardwired manner. Out of all homes, it is estimated that 60% of homes with a disposer have hardwiring and that the remaining 40% use a power outlet located in the sink cabinet. Notwithstanding these different home installation circumstances, conventional arrangements of operator-actuatable switching mechanisms for food waste disposers are typically configured to be wired in relation to the food waste disposers in a manner that is specifically tailored for implementation of the food waste disposer in relation to only a particular home installation circumstance, such as the circumstance when a Romex/BX cable is present.

For at least one or more of these reasons, or one or more other reasons, it would be advantageous if improved switch mechanisms or systems for use in or in conjunction with food waste disposers or other disposers could be developed, and/or if improved food waste disposers or other disposers having or operating in conjunction with such mechanisms or systems could be developed, and/or if improved methods of assembling and/or operating such mechanisms, systems, or disposers could be developed, so as to address any one or more of the concerns discussed above or to address one or more other concerns or provide one or more benefits.

BRIEF SUMMARY

In at least some example embodiments, the present disclosure relates to a food waste disposer system. The food waste disposer system includes a food waste disposer including a motor, a switch module coupled to the motor and operable to control power to the motor, and a housing including a bottom housing portion and a top housing portion, where the switch module and motor are supported within the housing. Additionally, the food waste disposer system also includes a cover switch mechanism including a cover control switch cord, a primary body having at least one actuator and configured to be coupled to the top housing portion, and a plurality of connecting components. The plurality of connecting components include a terminal cover configured to be coupled to the bottom housing portion and a switch interface connector coupled to the terminal cover and also configured to be coupled to the switch module. Also, the cover control switch cord extends between the primary body and the terminal cover, where a first end of the cover control switch cord extending into the terminal cover includes first and second wire leads, and where the first wire lead is coupled to the switch interface connector. Additionally, the terminal cover is further configured to receive a second end of a power link therewithin, where the switch interface connector is configured to be coupled to at least one third wire lead of the power link, and where the second wire lead is configured to be coupled at least indirectly to a fourth wire lead of the power link.

Further, in at least some further example embodiments, the present disclosure relates to a cover switch mechanism for implementation with a waste disposer having a motor, a switch module coupled at least indirectly to the motor, and a housing including a bottom housing portion and a top housing portion, wherein the switch module and motor are supported within the housing. The cover switch mechanism includes a cover control switch cord, a primary body configured to be coupled to the top housing portion, and a plurality of connecting components. The plurality of connecting components includes a terminal cover configured to be coupled to the bottom housing portion, a switch interface connector coupled to the terminal cover and also configured to be coupled to the switch module, and a wire lead connection component. The cover control switch cord extends between the primary body and the terminal cover, where a first end of the cover control switch cord extending into the terminal cover includes first and second wire leads, and where the first wire lead is coupled to the switch interface connector. Also, the terminal cover is further configured to receive a second end of a power link therewithin, where the switch interface connector is configured to be coupled to at least one third wire lead of the power link, and where the wire lead connection component is configured to couple the second wire lead with a fourth wire lead of the power link.

Additionally, in at least some additional example embodiments, the present disclosure relates to a method of installing a food waste disposer system. The method includes providing the food waste disposer system to an installation environment, where the food waste disposer system includes a housing, a switch module, and a motor, where the switch module and motor are supported at least indirectly upon the housing. Also, the method includes providing a cover switch mechanism including a cover control switch cord, a primary body configured to be coupled to the top housing portion, and a plurality of connecting components, where the plurality of connecting components include a terminal cover configured to be coupled to the bottom housing portion, a switch interface connector, and a wire lead connection component. Further, the method includes causing the cover control switch cord and a power link respectively to extend between a first region exterior of the terminal cover and a second region within the terminal cover by way of first and second orifices, respectively, of the terminal cover, so that each of respective first ends of the cover control switch cord and the power link extends into the second region. Additionally, the method includes coupling respective first wire leads provided at the respective first ends of the cover control switch cord and the power link to respective ports of the switch interface connector, and coupling together, at least indirectly, respective second wire leads provided at the respective first ends of the cover control switch cord and the power link, respectively. Further, the method includes assembling a terminal assembly including the terminal cover, the switch interface connector, and the first ends of the cover control switch cord and the power link in relation to a bottom portion of the housing, where the assembling includes coupling the switch interface connector to the switch module, and connecting the primary body of the cover switch mechanism to a top portion of the housing.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of food waste disposer systems (or other waste disposer systems), food waste disposers, and/or systems (or subsystems) employed in or in conjunction with such waste disposer systems/waste disposers, and/or related methods, are disclosed with reference to the accompanying drawings and are for illustrative purposes only. The systems and methods encompassed herein are not limited in their applications to the details of construction, arrangements of components, or other aspects or features illustrated in the drawings, but rather such systems and methods encompassed herein include other embodiments or are capable of being practiced or carried out in other various ways. Like reference numerals are used to indicate like components. In the drawings:

FIG. **1** is a bottom, substantially rear perspective view of a first example food waste disposer assembly including a first example food waste disposer assembled in combination with a first example cover control system, as can be installed in relation to another structure such as a sink, shown to be connected to a Romex/BX cable (shown in cutaway);

FIG. **2** is a top, substantially front perspective view of the first example food waste disposer assembly of FIG. **1**, again shown to be connected to the Romex/BX cable (shown in cutaway);

FIG. **3** is a bottom perspective view of the first example food waste disposer assembly and Romex/BX cable of FIGS. **1** and **2**, with each of a cover control switch cord of the assembly and the Romex/BX cable shown in cutaway;

FIG. **4** is a first cutaway, enlarged bottom perspective view of portions of the first example food waste disposer assembly of FIGS. **1**, **2**, and **3**;

FIG. **5** is a second cutaway, enlarged bottom perspective view of portions of the first example food waste disposer assembly of FIGS. **1**, **2**, **3**, and **4**;

FIG. **6** is a bottom plan view of a terminal cover of the first example food waste disposer assembly of FIGS. **1**, **2**, **3**, **4**, and **5**;

FIG. **7** is a bottom perspective view of the terminal cover of FIG. **6**;

FIG. **8** is a top perspective view of several LEF connecting components, including the terminal cover of FIGS. **6** and **7**, of the food waste disposer assembly of FIGS. **1** and **2**, assembled in combination with a cover control switch cord (shown in cutaway) of the food waste disposer assembly and the Romex/BX cable (shown in cutaway) of FIGS. **1** and **2**;

FIG. **9** is a perspective view of a start switch interface connector of the food waste disposer assembly of FIGS. **1** and **2**;

FIGS. **10**, **11**, **12**, **13**, **14**, **15**, **16**, **19**, and **20** respectively are perspective cutaway views of portions of the food waste disposer assembly of FIGS. **1** through **8** that, together with FIGS. **9**, **17**, and **18**, are intended to illustrate, collectively, steps of a method of assembly or installation of the food waste disposer assembly of FIGS. **1** and **2**;

FIGS. **17** and **18** provide two additional cutaway views of the food waste disposer assembly of FIGS. **1** and **2** that further illustrate portions of that assembly after a step of the method of assembly/installation represented by FIG. **16** has been completed;

FIG. **21** is a perspective view of an alternate embodiment of a terminal assembly in which a wire nut is employed to couple wire leads in place of a wire joiner connector; and

FIG. **22** is a perspective view of portions of the food waste disposer assembly of FIGS. **1** through **8** substantially corresponding to the portions shown in FIG. **10**, when the food waste disposer assembly is being assembled in accordance with a method of assembly/installation so as to be coupled to a power cord rather than the Romex/BX cable.

DETAILED DESCRIPTION

The present disclosure relates to and encompasses waste disposer systems and assemblies, such as food waste disposer assemblies, which are configured to be controlled by cover control mechanisms or systems (or subsystems). More particularly, in at least some embodiments, the present disclosure relates to food waste disposer assemblies having food waste disposers and cover control mechanisms, where the food waste disposers include motor sections at bottom ends (or at first ends) of the first waste disposers, and where the cover control mechanisms are mounted atop top ends (or at second ends) of the food waste disposers. Such a cover control mechanism for example can be provided in the form of a cover control kit, by which the cover control mechanism can be easily assembled in relation to a food waste disposer by a consumer. Such a cover control kit can be sold separately from the food waste disposer as a kit, or included along with the disposer (e.g., within the disposer box) when the disposer/disposer assembly is purchased by a consumer.

Additionally, the present disclosure envisions waste disposer assemblies in which the disposer assemblies including the cover control mechanisms can be assembled to receive electric power in any of several manners. More particularly, in at least some embodiments, the present disclosure relates to food waste disposer assemblies in which the food waste disposers, and additionally the cover control mechanism, can be coupled to receive electric power by each (or any) of a power cord that can be plugged into a wall outlet or other power source, or a Romex cable, or a BX hard covered wire. In at least some such embodiments, coupling of the food waste disposer assemblies to a Romex cable or BX hard covered wire can be achieved through implementation of a Romex/Bx connector. Additionally, in at least some embodiments, the present disclosure relates to a cover control mechanism (e.g., in the form of a cover control kit) that may be added to any of a variety of multiple different types of disposers, such that a consumer may pick or select a model of disposer from any of those different types of disposers and then apply/implement the cover control mechanism in relation to that selected disposer. The cover control mechanism is designed to include a connector by which the cover control mechanism may plug into the disposer, for any of the different types of disposers, so as to simplify installation of the cover control mechanism relative to the disposer.

The present disclosure envisions embodiments in which the waste disposer assembly is implemented by either corded installation, in which a cord associated with the waste disposer assembly can be plugged into an electric power (e.g., wall) outlet at the installation site, or hardwired installation, such that electric power can be delivered to the waste disposer assembly via a Romex/BX cable that is present at the installation site. That is, the present disclosure envisions that a given waste disposer assembly can include, or be implemented in conjunction with, a system (or system components) that enables the waste disposer assembly to be installed in either (or both) of two different manners relative to a surrounding environment such as a home installation site so that the given waste disposer assembly can be provided with electric power either by way of an electric power outlet or by way of a Romex/BX cable. Relatedly, the present disclosure envisions methods of installation of waste disposer assemblies according to which a waste disposer assembly can be installed in relation to a surrounding environment in either of two manners depending upon whether the waste disposer assembly is to receive electric power from an electric power outlet or via a Romex/BX cable.

Further, in at least some embodiments encompassed herein, the present disclosure relates to a disposer assembly having a lower end frame (LEF) and start switch, and a cover control mechanism, which are configured to facilitate coupling of the cover control mechanism to the disposer as well as to a power source by way of any of a power cord or Romex or BX cord or cable. In some such embodiments, the disposer assembly includes an arrangement by which the leads from a trigger switch (e.g., of the cover control mechanism) may be coupled to the start switch connector and also to the power cord leads, all within the same general area. Further, in some such embodiments, all of the wires are routed and connected into a new connector that mates with the start switch, as part of the LEF assembly.

More particularly, in at least some such embodiments encompassed herein, the disposer assembly is arranged in a manner according to which the two main cords—that is, the cord or cable with the leads from the cover control mechanism and also the cord or cable intended to link the disposer assembly with a power source (e.g., a Romex cable, BX hard covered cable, or power cord suited for being plugged into a wall outlet)—come into a junction box at different angles so the two cords can have individual strain reliefs. A new (two conductor) connector makes it possible to join the power leads within this small junction box. The junction box and new connector can be considered to be part of the cover control mechanism rather than the disposer, and this new cover control mechanism (or cover control kit) design resolves the interface to the new LEF and start switch and frees the customer from only having only one disposer model choice.

Referring to FIGS. **1** and **2**, a bottom, substantially rear perspective view and a top, substantially front perspective view, respectively, of a food waste disposer system or assembly **10** are shown, in accordance with a first example embodiment encompassed herein. As illustrated, the food waste disposer assembly **10** includes a food waste disposer **100** having an enclosure **102**, a cylindrical stator band **104**, and a lower end frame (LEF) **106** (see FIG. **1**). In general, the food waste disposer **100** can be understood as including a food conveying section, a motor section, and a grinding section. The food conveying section is generally positioned at a location corresponding to the location of the enclosure **102**, at or near the top of the food waste disposer **100**. The motor section is generally positioned at a location corresponding to and within the stator band **104**, at or near the bottom of the food waste disposer **100**. The grinding section is disposed between the food conveying section and the motor section. It should be appreciated that the food conveying section includes an inlet for receiving food waste and fluid (e.g., water), and conveys the food waste to the grinding section. The motor section includes a motor **170** imparting rotational movement to a motor shaft to operate the grinding section. In the present example embodiment, the motor **170** can be an electric motor that is an inductive motor, although the present disclosure is intended to encompass embodiments of food waste disposers employing other types of motors such as permanent magnet motors.

In the present example embodiment, the food waste disposer assembly **10** also includes, in addition to the food waste disposer **100**, a control switch mechanism **50**. The control switch mechanism **50** is configured to allow for users to turn on, turn off, or otherwise actuate or control operation of the disposer in relation to which the control switch mechanism is implemented. In the present example, the control switch mechanism **50** is considered to be distinct from the food waste disposer **100**, with the food waste disposer and control switch mechanism both being included as part of the food waste disposer assembly **10**. However, in other embodiments or contexts, the control switch mechanism **50** can instead be considered to form a part of the food waste disposer **100** itself. The control switch mechanism **50** can be included with the food waste disposer **100** at the time of purchase of that food waste disposer **100** (or of the overall food waste disposer assembly **10**), or can be obtained or purchased separately from the food waste disposer, as a cover control kit.

As shown, in the present embodiment, the control switch mechanism **50** includes a primary body **52**, LEF connecting components **54** (see FIG. **1**), and a cover control switch cord (or cable or wire) **56** linking the primary body **52** with the LEF connecting components **54**. The primary body **52** is configured to be mounted atop or at the top end of the food waste disposer **100**, along the cover of the food waste disposer. Due to this arrangement of the primary body **52**, actuation switches (or other actuators) **53** of the control switch mechanism **50** provided in the primary body are more easily accessed by users/operators, and the control switch mechanism **50** can also be referred to as a cover control mechanism. The cover control switch cord **56** is connected to the primary body **52** of the control switch mechanism **50** arranged at the top end of the food waste disposer **100** (e.g., to an AV extension tube **59** at the top of the disposer), and is arranged to run downward from the primary body **52** along the outside of the food waste disposer and then to pass under the food waste disposer to the LEF connecting components **54**. Arranged in this manner, the cover control switch cord **56** not only allows for the LEF connecting components **54** to be coupled to the primary body **52**, but also allows for the LEF connecting components to be positioned along the LEF **106**.

As will be described in further detail below, the positioning of the LEF connecting components **54** along the LEF **106** both facilitates coupling of the actuation switches (or other actuators) **53** of the primary body **52** of the control switch mechanism **50** to the motor section by way of the cover control switch cord **56**, and also facilitates the coupling of those actuation switches and the motor section with a power source. In the present example embodiment shown in FIGS. **1** and **2**, the LEF connecting components **54** are shown to be coupled to a Romex/BX cable **58**, by which power is received from a power source. The Romex/BX cable **58** is intended to be representative of either a Romex cable or a BX hard covered wire, as can be preinstalled in some homes or facilities for delivery of power to a device such as the food waste disposer assembly **10**. For purposes of the present description, the Romex/BX cable **58** is not considered to be a part of the food waste disposer assembly **10** although, in other contexts, the Romex/BX cable (or a portion of it, such as the portion nearby or coupled to the LEF connecting components **54**) can be considered to be a part of the food waste disposer assembly.

Although the example embodiment of FIGS. **1** and **2** illustrates the LEF connecting components **54** as being coupled to the Romex/BX cable **58**, the present example embodiment also is one that can accommodate implementation of a power cord having a plug that plugs into a wall outlet (e.g., a NEMA 5-15 plug), as will be additionally described below. That is to say, in the present embodiment, the LEF connecting components **54**, and the food waste disposer assembly **10** overall, are configured to allow for either implementations in which a Romex/BX cable **58** is coupled to the LEF connecting components **54** to allow for the supplying of power to the food waste disposer assembly, or implementations in which a power cord **400** (see FIG. **22**) is implemented in regard to the LEF connecting components **54** such that, upon the power cord also being plugged into a wall outlet, power can be supplied to the food waste disposer assembly **10**. Again, for purposes of the present description, such a power cord (e.g., the power cord **400**) is not considered to be a part of the food waste disposer assembly although, in other contexts, the power cord (or a portion of it, such as the portion nearby or coupled to the LEF connecting components **54**) can be considered to be a part of the food waste disposer assembly.

Turning to FIG. **3**, FIG. **4**, and FIG. **5**, first, second, and third bottom perspective views are provided of the food waste disposer assembly **10** or portions thereof, to more fully illustrate how a terminal cover (or junction box) **300** of the LEF connecting components **54** receives the cover control switch cord **56** as well as the Romex/BX cable **58**. More particularly, FIG. **3** is a bottom perspective view of the food waste disposer assembly **10** of FIGS. **1** and **2**, with each of the cover control switch cord **56** and the Romex/BX cable **58** shown in cutaway. By comparison, FIG. **4** is a first cutaway, enlarged bottom perspective view of portions of the food waste disposer assembly **10**, and FIG. **5** is a second cutaway, enlarged bottom perspective view of portions of the food waste disposer assembly **10**. FIGS. **3**, **4**, and **5** particularly show that the terminal cover **300** of the LEF connecting components **54** is coupled along a bottom surface **108** of the LEF **106** and is configured so that the cover control switch cord **56** enters the terminal cover at a first side wall **302** but the Romex/BX cable **58** enters the terminal cover at a bottom wall **304**. The bottom wall **304** is substantially parallel to the bottom surface **108** of the LEF **106**, and so the Romex/BX cable **58** enters the terminal cover **300** in a direction that is substantially normal (or perpendicular) to the bottom surface **108** of the LEF. By comparison, the cover control switch cord **56** enters the terminal cover **300** in a direction that is substantially parallel to or along the bottom surface **108** of the LEF **106**.

More particularly with respect to the terminal cover **300**, FIG. **6** provides a bottom plan view of the terminal cover and FIG. **7** provides a bottom perspective view of the terminal cover. As shown, the terminal cover (or junction box) **300** takes the form of a cup that includes, in addition to the bottom wall **304**, the first side wall **302** that extends upward from the bottom wall (downward when viewed as shown in FIG. **7**) to a first rim portion **314**, and an additional curved side wall **306** that also extends upward from the bottom wall to a second rim portion **316**. The bottom wall **304** includes a bottom orifice **320** that is configured to receive the Romex/BX cable **58** and the first side wall **302** includes a side orifice **322** that is configured to receive the cover control switch cord **56**. The additional curved side wall **306** extends around the bottom wall **304** from a first upwardly-extending edge **308** of the first side wall **302** to a second upwardly-extending edge **310** of the first side wall, such that the bottom wall **304**, first side wall **302**, and additional curved side wall **306** together define the cup in terms of defining an interior space **312** (see also FIG. **8**). The interior space **312** is entirely enclosed except for the bottom orifice **320**, the side orifice **322**, and a top opening **324** defined by the first rim portion **314** and the second rim portion **316**.

In addition, the terminal cover **300** includes a first rim extension (or lip) **326** and a second rim extension (or lip) **328**, each of which extends outward from the second rim portion **316** parallel or substantially parallel to the bottom wall **304** and can be considered to form a part of second rim portion. More particularly, the first rim extension **326** is a substantially triangular formation that extends generally outward away from a first side portion **330** of the additional curved side wall **306** that is adjacent to the first upwardly-extending edge **308**. Also, the second rim extension **328** is a substantially rectangular formation that extends generally outward away from a second side portion **332** of the additional curved side wall **306** that is adjacent to the second upwardly-extending edge **310**. The first rim extension **326** includes a round orifice **334** therein and the second rim extension **328** includes an elongated slot **336** that also extends generally outward away from the second side portion **332**.

In the present embodiment, the terminal cover **300** is made as a zinc casting. The use of a zinc casting can be desirable in terms of making it easier to achieve the complex shape of the terminal cover **300** that is suitable for the desired manner of wire routing. In other embodiments, the terminal cover **300** can be made from other materials such as steel, or in other manners.

Turning to FIG. **8**, the LEF connecting components **54** in the present embodiment include not only the terminal cover **300** but also a strain relief component (or grommet) **340**, an additional strain relief component (or simply strain relief) **342**, and a start switch interface connector **344** (see FIG. **9**). Also, as will be described further below, the LEF connecting components **54** in the present embodiment additionally include a wire joiner or wire joiner connector **346** (see FIG. **13**), although in another embodiment a wire nut **348** is employed in place of the wire joiner connector. The cover control switch cord **56** can be affixed to the terminal cover **300** by way of the strain relief component **340**, and the Romex/BX cable **58** can be secured in relation to the terminal cover **300** by way of the strain relief **342**. The strain relief **342** (which can also be referred to as a power link coupler, or a Romex/BX coupler) in the present embodiment can take the form of a commonly available electrical component typically used to ground metallic sheathed cable (BX) and/or secure the Romex/BX cable to another component such as, for example, an electrical outlet box or an adapter (in alternate embodiments, the strain relief can take other forms, such as a grommet).

Further, the LEF connecting components **54** are configured to facilitate the coupling of first and second wire leads **350** and **352**, respectively, of the cover control switch cord **56**, and third, fourth, and fifth wire leads **354**, **356**, and **357**, respectively (the white/neutral, green/ground, and black/live leads, respectively), of the Romex/BX cable **58**, with one another and with the start switch interface connector **344** as described in further detail below with respect to FIGS. **9** through **20**. Additionally, the start switch interface connector **344** is configured to be coupled to a start switch **388** (see FIG. **16**) of the food waste disposer **100** as also described in further detail below.

Turning to FIGS. **9-20**, additional perspective views are shown of the food waste disposer assembly **10** and portions thereof, particularly to illustrate a method of assembly or installation of the food waste disposer assembly in relation to the Romex/BX cable **58**, in accordance with an example embodiment encompassed herein. The method can be performed by any of a variety of operators or users, such as a customer who purchased the food waste disposer **100**, or a professional technician or installer. In particular, the method represented by FIGS. **9-20** in the present embodiment is one in which the control switch mechanism **50**, as can be provided in the form of a cover control kit (or as can be provided along with the food waste disposer **100** when purchase), is assembled in relation to the food waste disposer **100**. Also, in the method represented by FIGS. **9-20**, the Romex/BX cable **58** is coupled to the control switch mechanism **50** and to the food waste disposer **100**. By way of the method, the food waste disposer assembly **10** is coupled to a power source by way of the Romex/BX cable **58** and also is configured to allow operator control of the food waste disposer by the control switch mechanism **50**. Although the method as shown by FIGS. **9-20** entails coupling of the food waste disposer assembly **10** to a power source by way of the Romex/BX cable **58**, the present disclosure also encompasses additional methods that entail coupling the food waste disposer assembly to a power source by way of a power cord, as discussed further below.

In the present embodiment, the method of FIGS. **9-20** begins at a first step represented by FIG. **9**, which shows a perspective view of the start switch interface connector **344**. In the first step represented by FIG. **9**, an operator strips (e.g., by way of a wire stripping tool) each of the third and fourth wire leads **354** and **356** (e.g., white and green wires) of the Romex/BX cable **58** and the first wire lead **350** (or alternatively the second wire lead **352**) of the cover control switch cord **56**. In the present embodiment, each of these wire leads **354**, **356**, and **350** (or **352**) is stripped to a push in terminal distance. As illustrated in FIG. **9**, the start switch interface connector **344** includes a wire stripping feature **358** that shows the extent to which each of the third, fourth, and first wire leads **354**, **356**, and **350** (or the second lead **352**) should be stripped.

Next, at a second step represented by FIG. **10**, the operator inserts and secures, relative to the terminal cover **300**, each of the cover control switch cord **56** and the Romex/BX cable **58**. More particularly, in this second step, the operator inserts the strain relief component (or grommet) **340** into the side orifice **322** of the terminal cover **300** and additionally inserts the cover control switch cord **56** (the end opposite the end coupled to the primary body **52**) including the first and second wire leads **350** and **352** through the side orifice **322** and through the strain relief component therewithin, from the exterior of the terminal cover **300** into the interior space **312** of the terminal cover. Additionally, the operator inserts the strain relief **342** into (or in relation to) the bottom orifice **320** of the terminal cover **300** and additionally inserts the Romex/BX cable **58** (the free end opposite the end coupled to any power source) including the third, fourth, and fifth wire leads **354**, **356**, and **357** through the bottom orifice **320** and through the strain relief (or grommet) **342**, from the exterior of the terminal cover **300** into the interior space **312** of the terminal cover. Upon completion of this second step, each of the cover control switch cord **56** and the Romex/BX cable **58** are coupled to and secured relative to the terminal cover **300**, in a manner such that each of the first, second, third, fourth, and fifth wire leads **350**, **352**, **354**, **356** and **357** are positioned in the interior space **312**, as shown in FIG. **8**.

Next, at a third step represented by FIG. **11**, the operator assembles/couples the start switch interface connector **344** with respect to each of the cover control switch cord **56** and the Romex/BX cable **58**. To achieve this coupling, the operator pushes in the third and fourth wire leads **354** and **356** of the Romex/BX cable **58** (which were stripped in the first step) into first and second input ports **360** and **362**, respectively, of the start switch interface connector **344**. Additionally, the operator pushes the first wire lead **350** (or alternatively the second wire lead **352**, depending upon whether the first or second wire lead was stripped in the first step) into a third input port **364** of the start switch interface connector **344**. Further, at a fourth step represented by FIG. **12**, the operator snaps the start switch interface connector **344** into the terminal cover **300** so as to be secured relative to the terminal cover. This snapping of the start switch interface connector **344** in relation to the terminal cover **300** in the present embodiment is achieved by positioning pronged deformable snapping features **366** of the start switch interface connector (see also FIG. **9**) into and at least partly through the elongated slot **336** until the snapping features snap into place behind the elongated slot as shown in FIG. **12**. FIG. **12** particularly illustrates the combination of the start switch interface connector **344** and the terminal cover **300** when assembled in this manner, and with the snapping features **366** jutting out of the elongated slot **336** after having been pushed therethrough.

Next, at a fifth step shown in FIG. **13**, the operator couples the second wire lead **352** of the cover control switch cord **56** to the fifth (e.g., black) wire lead **357** of the Romex/BX cable **58**, by way of the wire joiner connector **346**. Upon joining the second and fifth wire leads **352** and **357**, an overall terminal assembly **380** including the terminal cover **300**, the start switch interface connector **344**, the strain relief component **340**, the power link coupler **342**, and the wire leads **350**, **352**, **354**, **356**, and **357** and associated ends of the cover control switch cord **56** and the Romex/BX cable **58**, is ready to be installed or coupled in relation to the food waste disposer **100** and particularly the LEF **106**. For purposes of the present description, it will be appreciated that the terminal assembly **380** both includes components that are considered parts of the food waste disposer assembly **10**, such as the LEF connecting components **54** including the terminal cover **300**, and also includes components that are (at least as described above) not considered to be part of the food waste disposer assembly **10**, namely, the third, fourth, and fifth wire leads **354**, **356**, and **357** of the

Romex/BX cable **58**. Nevertheless, in other contexts, the terminal assembly can be understood to include all of the components of the terminal assembly **380** except for any portions of the Romex/BX cable **58** such as the wire leads **354**, **356**, and **357**.

Installation/assembly of the terminal assembly **380** in relation to the LEF **106** generally involves three steps. At a sixth step represented by FIGS. **14** and **15**, the terminal assembly **380** is aligned and fitted into the LEF **106**. More particularly in this regard, FIG. **14** provides a bottom perspective view of portions of the LEF **106** in relation to the terminal assembly **380**, as the terminal assembly is approaching the bottom surface **108** of the LEF **106**. As illustrated, the start switch interface connector **344** includes first and second keying tabs (or protrusions) **382** arranged on opposite sides of that start switch interface connector, generally above opposite sides of the second rim extension **328**. The LEF **106** includes a generally-rectangular LEF opening (or orifice) **384** that includes first and second keying recesses **386** (see also FIG. **15**) that respectively are complementary in shape to the first and second keying tabs **382**, such that the start switch interface connector is keyed to the LEF opening. The LEF opening **384** including the keying recesses **386** is shaped and sized so that, as the terminal assembly **380** reaches the bottom surface **108** of the LEF **106**, the start switch interface connector **344** is able to proceed into the food waste disposer **100** and be positioned above an inner (top or inwardly-facing) surface **383** of the LEF, as shown in FIG. **15**, which provides a top plan view of portions of that inner surface and the terminal assembly **380**. Although the start switch interface connector **344** is positioned above the inner surface **383** of the LEF **106** when the terminal assembly **380** is aligned and fitted into the LEF **106**, it should further be appreciated from FIGS. **14** and **15** that the terminal cover **300** remains positioned entirely below the bottom surface **108** of the LEF **106**, with the first and second rim extensions **326** and **328** being positioned so as to abut and interface that bottom surface.

Additionally, subsequent to the sixth step, at a seventh step represented by FIG. **16** the terminal assembly **380** is slid relative to the LEF **106** so that the start switch interface connector **344** is pushed ahead tight against a start switch **388** that is mounted and supported upon the inner surface **383** of the LEF **106**. It will be appreciated also from FIG. **15** that the start switch interface connector **344** includes first, second, and third output ports **390**, **392**, and **394**, respectively, that are opposed to (and coupled electrically to) the first, second, and third input ports **360**, **362**, and **364**, respectively, and additionally that the start switch **388** includes first, second, and third input tabs **396**, **398**, and **397**, respectively. Further there is a gap (not shown) at least between certain portions of the start switch interface connector **344** such as the keying tabs **382** and certain portions of the terminal cover **300** such as the first and second rim extensions **326** and **328**. Given this arrangement, it is possible for a user/installer to cause the terminal assembly **380**, and particularly the start switch interface connector **344**, to slide in a direction indicated by an arrow **389** shown in FIG. **16** relative to the LEF **106** until (as shown in FIG. **16**) the respective first, second, and third input tabs **396**, **398**, and **397** are positioned within, and electrically coupled to, the respective first, second, and third output ports **390**, **392**, and **394**, respectively.

Given such installation, it will be appreciated that the first, second, and third input tabs **396**, **398**, and **397** of the start switch **388** are electrically coupled to the third, fourth, and first wire leads **354**, **356**, and **350**, respectively. FIGS. **17** and **18** respectively show first and second side perspective, cutaway views of the food waste disposer **100** and terminal assembly **380** when the terminal assembly has been slid relative to the LEF **106** so that the start switch interface connector **344** is coupled to the start switch **388**. Further, when such installation has been completed, it should be recognized that a switchable electrical circuit is completed among each of the Romex/BX cable **58**, the control switch mechanism **50**, and the start switch **388** (and associated motor of the food waste disposer). In particular, this electrical circuit can be understood to extend from the fifth wire lead (e.g., the black, live power lead) **357** of the Romex/BX cable **58** to the second wire lead **352** of cover control switch cord **56** coupled thereto by way of the wire joiner connector **346**. Further, this electrical circuit can be understood to extend additionally through the control switch mechanism **50**, from the second wire lead **352** via the cover control switch cord **56** to actuation switch(es) **53** within the primary body **52**, and back to the first wire lead **350** via the cover control switch cord. Additionally, this electrical circuit can be understood to extend further to the motor **170** via the start switch **388** and the start switch interface connector **344**, insofar as the first wire lead **350** is coupled to the start switch interface connector that is in turn connected to the motor via the switch module (depending upon switching status of the switch module). Further, the electrical circuit is completed, at least in terms of the circuit being coupled back to the Romex/BX cable **58**, insofar as the motor **170** is coupled to one or both of the third wire lead **354** and the fourth wire lead **356** of the Romex/BX cable again by way of the start switch **388** and start switch interface connector **344**.

Upon completion of the seventh step, an eighth step in the installation process is performed in which the terminal assembly **380** is secured to the food waste disposer **100** and particularly to the LEF **106**, as shown in FIG. **19**. In the present example embodiment, the terminal assembly **380** is secured to the food waste disposer **100** by way of screwing-in a fixing screw **309** through the round orifice **334** in the terminal cover **300** and then subsequently into a receiving orifice (not shown) that is provided in the LEF **106**, when the terminal assembly **380** has been slid into position (so that the start switch interface connector **344** is engaged with the start switch **388**) as described in regard to FIGS. **16**, **17**, and **18**). By screwing-in the fixing screw **309**, the terminal assembly **380** is locked into position relative to the LEF **106**. In particular, when locked in this position, the terminal assembly **380** cannot slide away from and disengage from the start switch **388**. Further, when locked in this position, the keying tabs **382** are not aligned with the keying recesses **386**, and thus the terminal assembly **380** is retained against the LEF **106** insofar as the start switch interface connector **344** is in contact with the inner surface **383** and cannot pass through the LEF **106** out of the food waste disposer **100**. That is, by virtue of the keying tabs (or LEF tabs) **382** being positioned along the inner surface **383** of the LEF **106** at locations where there are none of the keying recesses **386**, the terminal assembly **380** (and particularly the front of the terminal assembly, that is, the portion closest to the start switch **388**) also is secured relative to the LEF.

It should be appreciated that, depending upon the embodiment, the fixing screw **309** can be considered a part of the food waste disposer **100** (and/or the LEF **106** thereof) or alternatively can be considered one of the LEF connecting components **54** (and/or part of the terminal assembly **380**). In embodiments in which the fixing screw **309** is included as part of the food waste disposer **100** (e.g., as originally purchased), the above-described method should be understood to include a preliminary step of removing the fixing screw **309** from the LEF **106** prior to the installation of the terminal assembly **380** with respect to the LEF **106** (e.g., prior to the sixth step of FIGS. **14** and **15** as described above). Alternatively, if the fixing screw **309** is considered to be one of the LEF connecting components **54**, then no preliminary step of removing the fixing screw from the LEF needs to be performed.

Finally, after the terminal assembly **380** is secured to the LEF **106** by way of installation of the fixing screw **309**, a ninth step is performed as illustrated by FIG. **20**. In this ninth step, the primary body **52** of the control switch mechanism **50** (e.g., the cover control switch) is attached in place with respect to the AV extension tube **59** at the top of the food waste disposer **100**. In the present embodiment, the primary body **52** can be snapped in place by moving the primary body **52** relative to the AV extension tube **59** in a direction as indicated by an arrow **311**.

The present disclosure is intended to encompass numerous other embodiments and arrangements in addition to those described above with respect to FIGS. **1-20**. For example, although the above-described embodiment envisions that the LEF connecting components **54** include the wire joiner connector **346**, in an alternate embodiment the LEF connecting components can instead include a wire nut **399**. For such an embodiment employing a wire nut, the fifth step (previously described with respect to FIG. **13**) can take an alternative form in which the operator couples the second wire lead **352** of the cover control switch cord **56** to the fifth (e.g., black) wire lead **357** of the Romex/BX cable **58** by way of the wire nut **399**, as shown in FIG. **21**.

Additionally, as already noted above, the present disclosure relates to food waste disposer assemblies in which the food waste disposers, and additionally the cover control mechanisms, can be coupled to receive electric power by a power cord that can be plugged into a wall outlet or other power source, rather than coupled to receive electric power by way of the Romex/BX cable **58**. More particularly, although FIGS. **1** through **21** envision the food waste disposer assembly **10** as having the LEF connecting components **54** coupled to the Romex/BX cable **58**, the present disclosure also envisions embodiments of food waste disposer assemblies in which the LEF connecting components **54** are coupled to the power cord **400** (see FIG. **22**) that plugs into a wall outlet (not shown).

Further, it should be appreciated that the process described above with reference to FIGS. **9** through **20** can be performed not only to assemble the food waste disposer assembly **10** in relation to the Romex/BX cable **58**, but also to assemble the food waste disposer assembly in relation to the power cord **400**. That is, the process is identical except insofar as the steps involving coupling of the Romex/BX cable **58** in relation to the LEF connecting components **54** are modified to be steps involving coupling of the power cord **400** in relation to those LEF connecting components. In this regard, it should be appreciated that, in the modified version of the first step represented by FIG. **9**, an operator strips (e.g., by way of a wire stripping tool) each of the third and fourth wire leads of the power cord **400** in addition to the first wire lead **350** (or alternatively the second wire lead **352**) of the cover control switch cord **56**.

FIG. **22** particularly illustrates a modified version of the second step of FIG. **10** when that second step is performed in relation to the power cord **400** rather than the Romex/BX cable **58**. More particularly, in this modified version of the second step, the operator inserts the strain relief component (or grommet) **340** into the side orifice **322** of the terminal cover **300** and additionally inserts the cover control switch cord **56** (the end opposite the end coupled to the primary body **52**) including the first and second wire leads **350** and **352** through the side orifice **322** and through the strain relief component therewithin, from the exterior of the terminal cover **300** into the interior space **312** of the terminal cover. Additionally, the operator inserts the power link coupler **342** into (or in relation to) the bottom orifice **320** of the terminal cover **300** and additionally inserts the power cord **400** (the free end opposite the end intended for coupling to a wall outlet) including third, fourth, and fifth wire leads thereof (corresponding to the third, fourth, and fifth wire leads **354**, **356**, and **357** of the Romex/BX cable **58**) through the bottom orifice **320** and through the power link coupler (or grommet) **342**, from the exterior of the terminal cover **300** into the interior space **312** of the terminal cover. Upon completion of this modified version of the second step, each of the cover control switch cord **56** and the power cord **400** are coupled to and secured relative to the terminal cover **300**, in a manner such that each of the first and second wire leads **350**, **352** of the cover control switch cord and the third, fourth, and fifth wire leads of the power cord **400** are positioned within the interior space **312** (again see FIG. **8**).

It should further be appreciated that, subsequent to the performing of the modified version of the second step as illustrated by FIG. **22**, the method of assembling the food waste disposer assembly proceeds in the same manner as described above with respect to FIGS. **11** through **20**, except insofar as the method involves the power cord **400** (and associated third, fourth, and fifth wire leads) rather than the Romex/BX cable **58**. Thus, in contrast to the third, fourth, and fifth steps illustrated by FIGS. **11**, **12**, and **13**, respectively, the modified versions of those steps—and particularly the modified versions of the third and fifth steps—involve coupling of the third and fourth wire leads of the power cord **400** to the start switch interface connector **344** and the coupling of the fifth wire lead of the power cord to the second wire lead **352** of the cover control switch cord **56**, by way of the wire joiner connector **346** (or alternatively the wire nut **399** as mentioned in relation to FIG. **21**). Further, in contrast to the sixth step illustrated by FIGS. **14** and **15**, the seventh step illustrated by FIGS. **16**, **17**, and **18**, and the eighth step illustrated by FIG. **19**, the modified versions of those steps are identical except insofar as the terminal assembly that is aligned with and coupled to the LEF **106** includes a portion of the power cord **400** rather than the Romex/BX cable **58** that is included with the terminal assembly **380**. The method then again finishes with the ninth step illustrated by FIG. **20**, in which the primary body **52** of the control switch mechanism **50** (e.g., the cover control switch) is attached in place with respect to the AV extension tube at the top of the food waste disposer.

It will be appreciated from the above discussion that the LEF connecting components **54** allow for either the Romex/BX cable **58** or the power cord **400** to be coupled to the start switch **388** by way of the start switch interface connector **344** and wire joiner connector **346** (or wire nut **399**), depending upon the installation circumstance or environment. Also, in some installation circumstances or environment, it can be possible for a food waste disposer assembly to be coupled to a power source by way of either one of the Romex/BX cable **58** or the power cord **400**, if both the Romex/BX cable is present and also a wall outlet is present. Given these considerations, it will be appreciated that in at least some circumstances or embodiments, the methods of assembly/installation of the food waste disposer assembly described above in regard to FIGS. **9-22** can include a preliminary step (prior to the first step) at which the operator determines whether installation involving the Romex/BX cable **58** or installation involving the power cord **400** is appropriate taking into account the installation circumstance, environment, or operator preference (if both the Romex/BX cable **58** and wall outlet are present). If it is determined at this preliminary step that installation involving the Romex/BX cable **58** is appropriate, then the method continues as described above in regard to FIGS. **9** through **20** (and/or FIG. **21**, if the wire nut **399** is utilized instead of the wire joiner connector **346**). Alternatively, if at this preliminary step, it is determined that installation involving the power cord **400** is appropriate, then the method continues with the modified versions of the steps of the process described above in regard to FIGS. **9** through **20**, with those steps being modified to accommodate the power cord as shown in FIG. **22** (also, again, the wire joiner connector **346** or the wire nut **399** can be utilized depending upon the embodiment).

The present disclosure is also intended to encompass further embodiments and modified versions of the above-described embodiments in addition to the embodiments specifically described above. Among other things, although the above description relates to food waste disposers, the present disclosure is also intended to encompass embodiments relating to other types of waste disposers. Also, notwithstanding the description above regarding embodiments in which a terminal cover (or junction box) is attached to the LEF of the disposer and according to which the control/trigger switch cords and power cords enter the junction box at right angles or substantially right angles, the present disclosure is also intended to encompass other embodiments. For example, in an some alternate embodiments, both wire bundles (e.g., associated with the cover control mechanism and associated with the power cord or Romex or BX cord or cable) may be inserted straight up through the bottom of the disposer with a combined strain relief. Also, depending upon the embodiment, wire joiner connectors, standard wire nuts, or other connectors or fasteners can be employed to join the various conductors. Also, although the present disclosure envisions embodiments in which a food waste disposer assembly is coupled to a wall outlet by way of a power cord having a plug such as a NEMA 5-15 plug, the present disclosure is also intended to encompass other embodiments that include or operate in conjunction with other types of connectors, plugs, and adapters, including for example C-13 or C14 sockets or plugs.

It is specifically intended that the present invention not be limited to the embodiments and illustrations contained herein, but include modified forms of those embodiments including portions of the embodiments and combinations of elements of different embodiments as come within the scope of the following claims.

We claim:

1. A food waste disposer system comprising:

a food waste disposer including a motor, a switch module coupled to the motor and operable to control power to the motor, and a housing including a bottom housing portion and a top housing portion, wherein the switch module and motor are supported within the housing; and

a cover switch mechanism including a cover control switch cord, a primary body having at least one actuator and configured to be coupled to the top housing portion, and a plurality of connecting components,

wherein the plurality of connecting components include a terminal cover configured to be coupled to the bottom housing portion and a switch interface connector coupled to the terminal cover and also configured to be coupled to the switch module,

wherein the cover control switch cord extends between the primary body and the terminal cover, wherein a first end of the cover control switch cord extending into the terminal cover includes first and second wire leads, and wherein the first wire lead is coupled to the switch interface connector, and

wherein the terminal cover is further configured to receive a second end of a power link therewithin, wherein the switch interface connector is configured to be coupled to at least one third wire lead of the power link, and wherein the second wire lead is configured to be coupled at least indirectly to a fourth wire lead of the power link.

2. The food waste disposer system of claim **1**, wherein the power link is one of a Romex cable or a BX cable.

3. The food waste disposer system of claim **1**, wherein the power link is a power cord that extends from the second end to a third end at which the power cord includes a plug suitable for being coupled to a wall outlet.

4. The food waste disposer system of claim **1**, wherein the plurality of connecting components include a first wire lead connective component by which the second wire lead can be coupled to the fourth wire lead of the power link.

5. The food waste disposer system of claim **4**, wherein the first wire lead connective component is either a wire joiner connector or a wire nut.

6. The food waste disposer system of claim **1**, wherein the terminal cover is cup-shaped and has a bottom wall and at least one side wall extending upward from the bottom wall and configured to engage the bottom housing portion.

7. The food waste disposer system of claim **6**, wherein the terminal cover includes a first orifice within the bottom wall and a second orifice within the at least one side wall, wherein the cover control switch cord extends into the terminal cover by the first orifice, and wherein the second end of the power link is received by the terminal cover by the second orifice.

8. The food waste disposer of claim **7**, further comprising at least one strain relief component by which one or both of the cover control switch cord and the power link can be at least partly secured to the terminal cover, and wherein the bottom wall is substantially perpendicular to the at least one side wall.

9. The food waste disposer of claim **6**, wherein the bottom housing portion includes a first orifice having a first recessed indentation, wherein the switch interface connector includes a first protrusion that is substantially complementary in shape to the first orifice, wherein the switch interface connector is able to pass into an interior cavity within the food waste disposer when the first protrusion is aligned with the first recessed indentation, and wherein the first protrusion is no longer aligned with the first recessed indentation when the switch interface connector is coupled to the switch module, such that the switch interface connector is not able to pass out of the interior cavity when the switch interface connector is coupled to the switch module.

10. The food waste disposer of claim **9**, wherein the terminal cover includes a first rim extension extending from a rim of the at least one side wall positioned above the bottom wall, and further comprising a fixing screw that serves to couple the rim to the bottom housing portion and further serves to prevent the switch interface connector from being decoupled from the switch module.

11. The food waste disposer of claim 1, wherein the switch interface connector is configured to allow for electrical coupling between each of the first wire lead and the at least one third wire lead and respective terminals of the switch module, wherein a switchable electrical circuit is formed that extends between the second wire lead and the fourth wire lead as coupled by a first wire lead connective component, further extends between the second wire lead and the first wire lead by way of the cover switch mechanism, and additionally extends from the first wire lead to one or more of the at least one third wire lead at least indirectly by way of the switch interface connector, the switch module, and the motor.

12. A cover switch mechanism for implementation with a waste disposer having a motor, a switch module coupled at least indirectly to the motor, and a housing including a bottom housing portion and a top housing portion, wherein the switch module and motor are supported within the housing, the cover switch mechanism including:
- a cover control switch cord;
- a primary body configured to be coupled to the top housing portion; and
- a plurality of connecting components including:
  - a terminal cover configured to be coupled to the bottom housing portion;
  - a switch interface connector coupled to the terminal cover and also configured to be coupled to the switch module; and
  - a wire lead connection component,
- wherein the cover control switch cord extends between the primary body and the terminal cover, wherein a first end of the cover control switch cord extending into the terminal cover includes first and second wire leads, and wherein the first wire lead is coupled to the switch interface connector, and
- wherein the terminal cover is further configured to receive a second end of a power link therewithin, wherein the switch interface connector is configured to be coupled to at least one third wire lead of the power link, and wherein the wire lead connection component is configured to couple the second wire lead with a fourth wire lead of the power link.

13. The cover switch mechanism of claim 12, wherein the power link is one of a Romex cable or a BX cable.

14. The cover switch mechanism of claim 12, wherein the power link is a power cord that extends from the second end to a third end at which the power cord includes a plug suitable for being coupled to a wall outlet.

15. The cover switch mechanism of claim 12, wherein the first wire lead connective component is either a wire joiner connector or a wire nut, and the cover switch mechanism is provided as a kit that is supplemental to the food waste disposer.

16. The cover switch mechanism of claim 12, wherein the terminal cover is cup-shaped and has a bottom wall and at least one side wall extending upward from the bottom wall and configured to engage the bottom housing portion, wherein the terminal cover includes a first orifice within the bottom wall and a second orifice within the at least one side wall, wherein the cover control switch cord extends into the terminal cover by way of the first orifice, and wherein the second end of the power link is received by the terminal cover by way of the second orifice.

17. The cover switch mechanism of claim 16, wherein the plurality of connecting components additionally include at least one strain relief component, and wherein the bottom wall is substantially perpendicular to the at least one side wall.

18. A method of installing a food waste disposer system, the method comprising:
- providing the food waste disposer system to an installation environment, wherein the food waste disposer system includes a housing, a switch module, and a motor, wherein the switch module and motor are supported at least indirectly upon the housing;
- providing a cover switch mechanism including a cover control switch cord, a primary body configured to be coupled to the top housing portion, and a plurality of connecting components, wherein the plurality of connecting components include a terminal cover configured to be coupled to the bottom housing portion, a switch interface connector, and a wire lead connection component;
- causing the cover control switch cord and a power link respectively to extend between a first region exterior of the terminal cover and a second region within the terminal cover by way of first and second orifices, respectively, of the terminal cover, so that each of respective first ends of the cover control switch cord and the power link extends into the second region;
- coupling respective first wire leads provided at the respective first ends of the cover control switch cord and the power link to respective ports of the switch interface connector;
- coupling together, at least indirectly, respective second wire leads provided at the respective first ends of the cover control switch cord and the power link, respectively;
- assembling a terminal assembly including the terminal cover, the switch interface connector, and the first ends of the cover control switch cord and the power link in relation to a bottom portion of the housing, wherein the assembling includes coupling the switch interface connector to the switch module; and
- connecting the primary body of the cover switch mechanism to a top portion of the housing.

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